GenClasses
Your goal our support

QUICK POINT

UPPCS Prelims 2026 · Environment & Ecology · Pointer Notes

Complete Environment — from Ecology to Uttar Pradesh Special

Quick Revision Series

Environment & Ecology

2026 UPPCS

UPPCS Prelims Exam 2026

Every repeatedly-asked fact, made exam-ready — with maps, diagrams and updated data

103 Ramsar sites | **58** Tiger reserves | **13/18** UNESCO biospheres | **14** UP Ramsar sites

Visit our website for important Free Tests — Uttar Pradesh 2026

www.genclasses.com →

In this issue

1. Part 1: Ecology & Ecosystems
2. Part 2: Biodiversity & Conservation
3. Part 3: Pollution & Climate Change
4. Part 4: Conventions, Laws & Institutions
5. Part 5: Uttar Pradesh Special

www.genclasses.com | t.me/genclassesofficial | t.me/genclasses
genclassessupport@gmail.com | Help 9506321415

EXPERT'S REVIEW



Ritesh Patel

Range Forest Officer (RFO)

Uttar Pradesh Forest & Wildlife Department

✓ Recommends this booklet

> Working in the field with the Forest & Wildlife Department, I see every day that the environment is not just a chapter in a book — it is the reality on the ground. This GenClasses booklet puts that reality into the language of the exam: from the basic concepts of ecology to Dudhwa, the tiger reserves and Uttar Pradesh's Ramsar sites, every fact is precise and backed by the latest data.
>
> The maps, diagrams and 'Points to Note' make tough topics revisable in minutes. I recommend this booklet to every aspirant preparing for the UPPCS Prelims 2026.
>
> **— Ritesh Patel**
> Range Forest Officer, UP Forest & Wildlife Department

INSIDE THIS BOOKLET

Updated to Nov 2026
103 Ramsar, 58 tiger reserves, new counts

34 maps & diagrams
tough points made clear at a glance

Uttar Pradesh special
forests, sanctuaries, 14 Ramsar sites

Exam traps
Points to Note in every section

Recommended for UPPCS Prelims Exam 2026

Environment & Ecology — complete pointer notes

www.genclasses.com
t.me/genclassesofficial
t.me/genclasses
genclassessupport@gmail.com
Help 9506321415

A NEW ISSUE EVERY DAY

Booklets of every subject on Telegram

Quick revision booklets like this one — for every subject — keep coming daily and weekly on our Telegram channel.

Booklets for all subjects
Quick revision booklets for every subject, from History to Current Affairs

Daily & weekly updates
New booklets every day and every week, straight on Telegram

Extra Notes PDF
Additional notes on every topic, for revision before the exam

Concept Notes
Notes that explain tough topics in simple language

Free Tests — Uttar Pradesh 2026
www.genclasses.com

WhatsApp Support
9506321415

Booklets for all these subjects

History · Polity · Geography · Economy · Environment · Science · Art & Culture · Current Affairs · Uttar Pradesh Special

Join today — absolutely free

Be the first to know about new booklets, notes and free tests

t.me/genclassesofficial | t.me/genclasses

Telegram

Part 1: Ecology & Ecosystem

01 Ecology and its Basic Principles

PYQ 18×

Key Terms, Pioneers & Chronology

- **Term 'Ecology':** Greek Oikos (house) + Logos (study) • Reiter combined it as 'Oikology' • scientific definition & popularisation: German biologist **Ernst Haeckel (1866)** 5×
- **Father of Indian Ecology:** Prof. **Ramdeo Misra** • laid the foundation of ecology teaching & research at Banaras Hindu University 3×
- **Deep Ecology:** **Arne Naess** (Norway, 1973) • not human-centred but **nature-centred** (ecocentric) view 2×
- **Ecological Footprint:** **William Rees & Mathis Wackernagel** (1990s) • unit **global hectare** • measures human demand on ecosystems • basis of 'Earth Overshoot Day' 4×
- **Biosphere:** term by Austrian geologist **Eduard Suess (1875)** • ecological concept by **Vladimir Vernadsky (1926)**

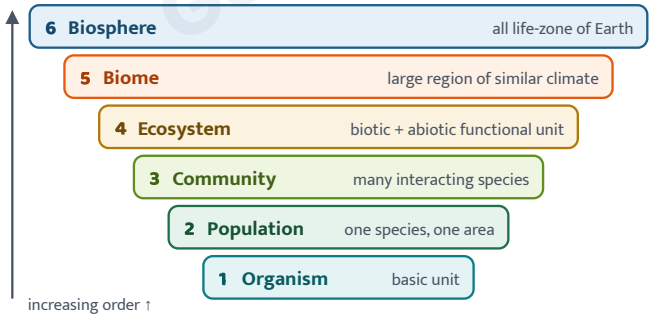
Points to Note

1. 'Ecology' → **Haeckel (1866)** • 'Ecosystem' → **Tansley (1935)** — both given together as options to confuse
2. Term 'Biosphere' is Suess's • the concept is Vernadsky's
3. Unit of ecological footprint is global hectare — not tonnes of CO_2 (that is carbon footprint)

Levels of Ecological Organisation 6×

- **Ascending order:** Organism → Population → Community → Ecosystem → Biome → Biosphere
- **Organism (individual):** smallest, basic unit of the hierarchy
- **Population:** individuals of the **same species** living in a defined area at the same time
- **Community:** **different species** (plants, animals, microbes) interacting in a defined area
- **Ecosystem:** functional unit of biotic community + abiotic environment
- **Biome:** large regional ecosystem with similar climate • tundra, taiga, savanna
- **Biosphere:** zone where land, water and air meet and life exists • from ≈8 km above sea level to ≈11 km ocean depth

Levels of Ecological Organisation



Points to Note

1. Population = **one** species • Community = **many** species
2. Community has only biotic components • add abiotic and it becomes an ecosystem
3. Largest level is biosphere • biome is larger than ecosystem

Habitat vs Ecological Niche 7×

- **Term 'Niche':** **Joseph Grinnell (1917)** • functional niche: Charles Elton • n-dimensional hypervolume: G. E. Hutchinson (1957)
- **Fundamental vs realised niche:** fundamental = full potential niche without competition • realised = part actually used after competition
- **Gause's Competitive Exclusion Principle (1934):** with limited resources, two species cannot coexist indefinitely in the same niche

Basis	Habitat	Niche
Concept	organism's **'address'** (physical place)	organism's **'profession'** (functional role)
Species	many species in one habitat (pond, forest)	one niche belongs to **one species** only
Components	only physical & climatic factors	place + position in food chain + breeding time + behaviour
Stability	relatively stable, broad	changes with life stage (tadpole ≠ adult frog)

Points to Note

1. 'Habitat = address, niche = profession' — E. P. Odum's analogy
2. Niches of two species are never fully identical • many species can share one habitat
3. Term 'niche' is **Grinnell's** — Elton and Hutchinson developed it

Ecotone, Edge Effect & Ecocline 5×

- **Ecotone:** transition zone between two or more different ecosystems • also 'tension zone'
- **Ecotone examples:** **mangroves** (land–sea) • **estuary** (fresh–salt water) • **grassland** (forest–desert) • **wetland** (water–land)
- **Edge Effect:** in an ecotone both number of certain species and population density are **higher** than in adjoining ecosystems
- **Edge Species:** found mainly/most abundantly in the ecotone • **birds** at edges of terrestrial ecosystems
- **Ecocline:** gradual, continuous change in temperature, salinity, humidity etc. from one ecosystem to another
- **Ecotype:** genetically adapted populations of the **same species** in different environments • can interbreed

Points to Note

1. Ecotone = zone • Ecocline = gradient
2. Edge effect means **higher** diversity & density — not lower
3. Ecotype is not a new species — interbreeding possible

Biotic Interactions 11×

- **Mutualism (+,+):** lichen (alga + fungus) • mycorrhiza (fungus + higher-plant roots) • Rhizobium–legumes • coral–zooxanthellae • pollinator–flower • fig–wasp

- **Commensalism (+,0):** **orchid (epiphyte)** on a tree branch • **remora** fish attached to whale/shark • barnacles on whale • **cattle egret** with grazing cattle • sea anemone–clownfish
- **Amensalism (–,0):** **Penicillium**'s penicillin → bacteria killed • **walnut (Juglans regia)** releases 'juglone' → nearby herbs fail to grow (**allelopathy**) • small plants stunted in shade of big trees
- **Parasitism (+,–):** draws nutrition from host, usually does not kill it at once • **Cuscuta (dodder)** = total stem parasite • **sandalwood** = partial root parasite • tapeworm, lice • koel laying eggs in crow's nest = **brood parasitism**
- **Predation (+,–):** predator kills and eats prey • lion–deer • insectivorous plants: **pitcher plant (Nepenthes)**, Drosera, Utricularia
- **Competition (–,–):** for limited food/space/light, efficiency of both falls • **flamingos vs resident fishes** for zooplankton in South American lakes

Points to Note

1. Orchid = **commensalism** (only support & light, no nutrition) • Cuscuta = **parasitism** (sucks sap)
2. Pitcher plant is autotrophic — eats insects only for **nitrogen**, as it grows in nitrogen-poor soil
3. Predation and parasitism are both (+,–) — predator kills, parasite usually does not
4. Walnut-juglone = amensalism (–,0), not competition

Major Ecological Rules 4×

- **Bergmann's Rule (1847):** **body size** of warm-blooded animals larger in cold regions, smaller in warm • less surface/volume → less heat loss
- **Allen's Rule (1877):** mammals of cold regions have shorter **appendages** (ears, tail, limbs) • Arctic fox vs desert (fennec) fox
- **Gloger's Rule:** warm-blooded animals of warm-humid regions have more **melanin** → darker colour
- **Jordan's Rule:** fishes of cold water have more **vertebrae**
- **Liebig's Law of the Minimum (1840):** growth limited by the essential element available in minimum quantity
- **Shelford's Law of Tolerance (1913):** each factor has a minimum–maximum tolerance limit • both deficiency and excess are limiting

Points to Note

1. Bergmann = **body size** • Allen = **appendages** — often swapped in options
2. Bergmann, Allen, Gloger = warm-blooded animals • Jordan = fishes

One-line Facts

- **"Ecology is permanent economy":** slogan of the **Chipko Movement** • popularised by **Sunderlal Bahuguna**
- **Loam soil:** best for plant growth • sand : silt : clay ≈ 40 : 40 : 20
- **Halophytes:** saline water/soil • negatively geotropic breathing roots **pneumatophores** • Sundari, Rhizophora, Avicennia • **vivipary** in mangroves
- **Xerophytes:** leaves modified into **spines** (less transpiration) • stem fleshy, green, photosynthetic • **Opuntia (prickly pear)**
- **Functional & structural unit of the biosphere:** **Ecosystem**

02 Ecosystem: Structure and Function

PYQ 24×

Concept & Classification

- **Term 'Ecosystem':** British ecologist **A. G. Tansley (1935)** 6×
- **Older synonymous terms:** Biocoenosis — Karl Möbius (1877) • Microcosm — Stephen Forbes (1887)
- **Nature:** **open system** — continuous input & output of energy and matter • Earth: nearly **closed** for matter, **open** for energy
- **Basic functions:** productivity • decomposition • energy flow • nutrient cycling
- **Largest & most stable ecosystem:** **ocean** • ≈71% of Earth 5×
- **Artificial ecosystems:** **paddy field** • wheat field • **aquarium** • garden • reservoir/dam • spacecraft 6×

Points to Note

1. Ecosystem is an **open** system — 'closed system' option is wrong • Earth open for energy, closed for matter
2. Paddy field, aquarium = artificial — shown as natural to trap

Components of an Ecosystem

- **Abiotic — physical:** light • temperature • humidity • air pressure • topography • wind
- **Abiotic — inorganic:** water • oxygen • CO2 • nitrogen • calcium • phosphorus • potassium
- **Abiotic — organic:** proteins • carbohydrates • fats • humus • urea • **link** between biotic and abiotic
- **Producers (autotrophs):** green plants • blue-green algae (cyanobacteria) • phytoplankton • **chemosynthetic bacteria**
- **Primary consumers (herbivores):** grasshopper • cow • goat • deer • rabbit • zooplankton • Elton: **'key-industry animals'** (plant tissue → animal tissue)
- **Secondary consumers (primary carnivores):** frog • fox • wolf • small fishes • lizard
- **Tertiary/top consumers:** lion • tiger • hawk • shark • peacock
- **Omnivores:** humans • bear • crow • cockroach • dog • ant (consumer + decomposer)
- **Decomposers (saprotrophs/micro-consumers):** mainly **bacteria** & **fungi** • enzymes break complex organic → simple inorganic 3×
- **Detritivores:** earthworm • millipede • dung beetle • termite • break detritus into small pieces (**fragmentation**)
- **Steps of decomposition:** fragmentation → leaching → catabolism → humification → mineralisation • fast in warm-moist conditions • slow in lignin/chitin-rich detritus

Points to Note

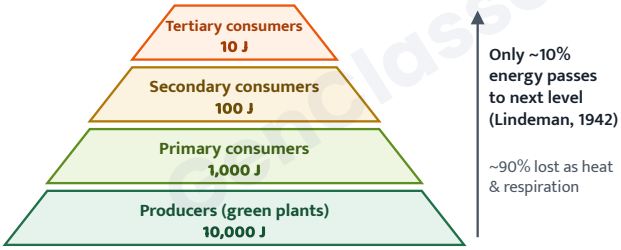
1. Viruses are **not** decomposers — inactive on dead matter
2. Earthworm is a detritivore (fragmenter) — true decomposers are bacteria & fungi
3. Chemosynthetic bacteria are also producers

Energy Flow & Laws of Thermodynamics 8×

- **Energy source:** **solar radiation** • **less than 50%** of incident radiation = Photosynthetically Active Radiation (**PAR, 400–700 nanometre**)
- **Use by plants:** only **2–10%** of PAR • ≈1–5% of total solar energy
- **Laws of thermodynamics:** first — energy only transformed, neither created nor destroyed • second — every transformation loses some energy as heat → less energy at each higher level
- **Energy flow:** always **unidirectional** • Sun → producers → herbivores → carnivores • never flows back 5×

- **Lindeman's 10% Law (1942):** **Raymond Lindeman** • only **10%** energy passes to next trophic level • **90%** lost in respiration, metabolism, heat • producer 1000 → 100 → 10 → 1 joule 4×
- **Result:** food chain usually has no more than **4–5 trophic levels**

Energy Flow & the 10% Law



Energy flow is always unidirectional · nutrients cycle

Points to Note

1. Energy flow is **unidirectional**, nutrients are **cyclic**
2. 10% law → **Lindeman (1942)** • pyramids → **Elton (1927)**
3. Plants capture 2–10% of PAR — not of total solar energy

Food Chain & Food Web

- **Grazing food chain:** starts from living green plants • directly depends on sunlight • main conduit of energy flow in **aquatic ecosystems**
- **Terrestrial example:** grass → grasshopper → frog → snake → hawk 4×
- **Marine example:** diatoms/phytoplankton → crustaceans/zooplankton → **herring** 2×
- **Detritus food chain:** starts from dead organic matter • in **terrestrial ecosystems** far more energy flows through it than through grazing chain • fallen leaves → earthworm/microbes → small bird → hawk
- **Food web:** interlocked network of many food chains • more complex web = more **stable** ecosystem

Points to Note

1. Aquatic: **grazing** chain dominant • terrestrial: more energy via **detritus** chain
2. The two chains are linked at some levels — not wholly separate

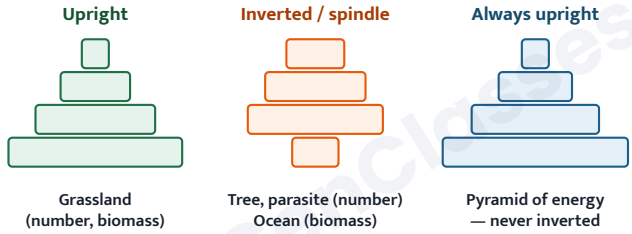
Ecological Pyramids 9×

- **Proposed by: Charles Elton (1927)** • 'Eltonian pyramids' • base always **producers**, apex top consumer

Pyramid	Upright	Inverted
Number	grassland, pond, crop field	a single large tree (parasitic chain)
Biomass	most terrestrial (forest, grassland)	pond/ocean
Energy	**always**	**never**

- **Pyramid of numbers of a tree:** tree (1) → herbivorous birds (many) → parasites (thousands) → hyperparasites (lakhs) = inverted • predatory birds in place of parasites = **spindle-shaped**
- **Biomass pyramid of ocean (inverted):** **standing crop** of phytoplankton less than that of fishes/whales • short life cycle, rapid reproduction 5×
- **Energy pyramid (always upright):** result of 10% law & second law of thermodynamics 4×
- **Limitations (besides decomposers):** does not show one species at several trophic levels • assumes simple chain, not food web

Ecological Pyramids: Upright vs Inverted



Producers at base · top consumers at apex

Points to Note

1. Energy pyramid is **always upright** — never inverted
2. Biomass: forest/grassland **upright**, pond/ocean **inverted**
3. Single tree: pyramid of **numbers** inverted, but pyramid of **biomass** upright

Ecological Productivity 4×

- **Gross Primary Productivity (GPP):** total rate of organic matter production in photosynthesis
- **Net Primary Productivity (NPP):** **NPP = GPP − R** (plant respiration) • this is what is available to herbivores & decomposers
- **Secondary productivity:** rate of formation of new organic matter by consumers
- **Highest NPP per unit area:** coral reefs • estuaries • swamps/marshes • **tropical rainforest** (highest among terrestrial)
- **Terrestrial descending order:** tropical rainforest > temperate forest > savanna > grassland > tundra/desert (lowest)
- **Open ocean:** very low per unit area • large total output due to vast area
- **Global NPP:** ≈ **170 billion tonnes**/year (dry weight) • oceans (≈70% of surface) only ≈ **55 billion tonnes**

Points to Note

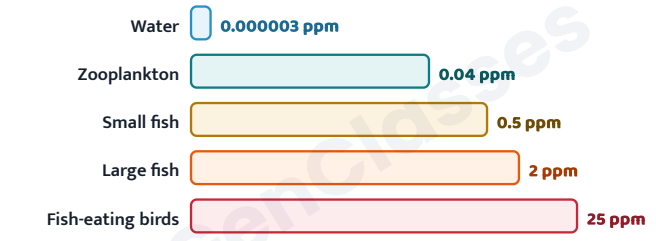
1. Herbivores get **NPP**, not GPP
2. Open ocean is poorly productive per unit area — 'most productive' option is wrong
3. Oceans cover ≈70% of surface yet give only ≈ one-third of global NPP

Bioaccumulation & Biomagnification 6×

- **Bioaccumulation:** build-up of a non-degradable pollutant in tissues of **one organism** over time • within one trophic level
- **Biomagnification:** progressive rise in toxin concentration towards **higher trophic levels**
- **Conditions:** **persistent** • **fat-soluble** (water-soluble ones are excreted in urine) • **biologically active**
- **DDT:** water 0.003 ppb → zooplankton 0.04 ppm → small fish 0.5 → large fish 2 → fish-eating birds 25 ppm • disturbs **calcium metabolism** → thin eggshells, premature breaking
- **Mercury (methylmercury):** fish to humans → **Minamata disease** (Japan)
- **Cadmium:** **Itai-itai disease** (Japan)
- **Diclofenac (veterinary painkiller):** vultures eating carcasses suffer **kidney failure** • >95% vultures wiped out • veterinary use banned 2006 • alternatives **meloxicam**, tolfenamic acid • aceclofenac, ketoprofen (veterinary use) also banned in 2023 4×

- **Others:** endosulfan • PCBs (polychlorinated biphenyls) • strontium-90

Biomagnification: the DDT Example



DDT concentration (log scale) rises with trophic level
Result: thin egg-shells in birds · also mercury, lead, cadmium

Points to Note

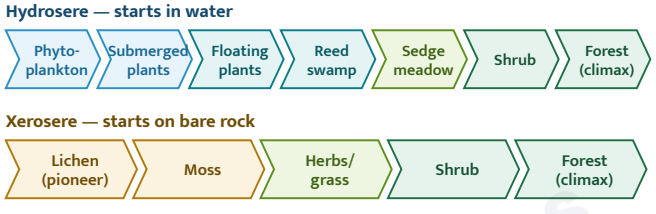
1. Bioaccumulation = within **one organism** • biomagnification = across **trophic levels**
2. Water-soluble pollutants do not biomagnify
3. Highest concentration in the **top consumer**
4. Mercury → Minamata • cadmium → Itai-itai — often swapped

Ecological Succession 5×

- **Definition:** gradual, directional, predictable change in species composition of a community over time • term **Hult (1885)** • theory **F. E. Clements (1916)**
- **Pioneer community:** first to colonise a bare area • rock (lithosere/xerosere): **lichens** (crustose) → mosses • water (hydrosere): **phytoplankton**
- **Climax community:** final, stable, in equilibrium with environment • dense forest
- **Sere:** sequence of successive stages from pioneer to climax
- **Clements' 5 stages:** **Nudation** (bare area) → **Invasion** (migration + ecesis + aggregation) → **Competition & coaction** → **Reaction** (change in soil & microclimate) → **Stabilisation** (climax) 3×
- **Hydrosere:** phytoplankton → submerged → floating → reed-swamp → sedge-meadow → shrub → forest

Basis	Primary succession	Secondary succession
Area	no earlier life/soil: bare rock, cooled lava, new sand dunes	community destroyed by flood, forest fire, deforestation
Soil	absent	present
Speed	very **slow**	**fast**

Stages of Ecological Succession



Both end in a climax community · primary vs secondary succession

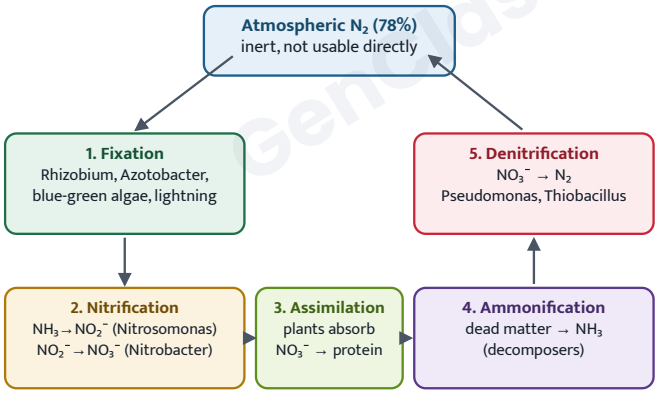
Points to Note

1. Pioneer on rock is **lichen** — moss comes second • in water **phytoplankton**
2. Both hydrosere and xerosere end in a **mesic** climax
3. Correct order: Nudation → Invasion → Competition → Reaction → Stabilisation

Biogeochemical Cycles 7×

- **Gaseous cycles:** reservoir **atmosphere/hydrosphere** • water • carbon • oxygen • **nitrogen** • fast, complete
- **Sedimentary cycles:** reservoir **Earth's crust/rocks** • **phosphorus** • **sulphur** • calcium • iron • slow
- **Free-living N-fixers:** **Azotobacter** (aerobic) • **Beijerinckia** (aerobic) • **Clostridium** (anaerobic) • **Rhodospirillum** • Nostoc, Anabaena (cyanobacteria)
- **Symbiotic N-fixers:** **Rhizobium** (legumes) • **Frankia** (non-legumes: Alnus, Casuarina) • **Anabaena azollae** (in Azolla fern — biofertiliser in paddy) • enzyme nitrogenase
- **Ammonification:** dead organic nitrogen → ammonia (decomposers)
- **Nitrification:** ammonia → nitrite: **Nitrosomonas, Nitrococcus** • nitrite → nitrate: **Nitrobacter**
- **Denitrification:** nitrate → atmospheric nitrogen • **Pseudomonas, Thiobacillus**
- **Phosphorus & carbon:** phosphorus cycle has **no gaseous phase** • ≈71% of total carbon dissolved in oceans

The Nitrogen Cycle



Points to Note

1. Plants take nitrogen mainly as **nitrate** (some ammonium) — not the 78% atmospheric N2 directly
2. Frankia = **non-leguminous** • Rhizobium = leguminous
3. Nitrosomonas first (nitrite), Nitrobacter later (nitrate) — order is reversed to trap
4. Phosphorus, sulphur = **sedimentary** • nitrogen, carbon = **gaseous**

One-line Facts

- **Main primary producers of the sea:** **phytoplankton** & diatoms • >50% of Earth's total oxygen
- **Biggest limitation of Eltonian pyramids:** no place for **decomposers** & saprophytes
- **Eutrophication:** excess nitrate-phosphate → **algal bloom** → dissolved oxygen (DO) falls, **BOD** rises 5×
- **Best indicators of water pollution:** **DO** & **BOD (Biological Oxygen Demand)** • highest BOD in the Ganga between **Kanpur–Prayagraj** 3×
- **Bionics:** science of building engineering/technical systems on principles of biological systems
- **Eucalyptus:** **'ecological terrorist'** • heavy groundwater uptake & transpiration • destroys soil fertility • native to **Australia** 4×

03 Major Biomes of the World & India and Aquatic Ecosystems

PYQ 16×

Major Terrestrial Biomes

- **Tundra:** polar regions & high mountain peaks • extremely cold, scanty precipitation (as snow) • permanently frozen subsoil (**permafrost**) • **treeless** — lichens, mosses, sedges, dwarf shrubs
- **Tundra — fauna:** reindeer/caribou, polar bear, arctic fox, lemming, musk ox • reptiles & amphibians almost absent • lowest biodiversity
- **Taiga / Boreal forest:** about 50°–65° N (Eurasia, North America) • long severe winter, short summer • **podzol** soil (acidic, mineral-poor)
- **Taiga — flora & fauna: coniferous evergreen** — pine, spruce, fir, larch • needle-shaped leaves • Siberian tiger, moose, lynx, mink, fox • **world's largest terrestrial biome** • softwood → paper & pulp industry
- **Temperate deciduous forest:** four distinct seasons • moderate temperature • rainfall 75–150 cm, year-round • oak, maple, beech, birch, elm — shed leaves before winter • bear, deer, raccoon, wolf • humus-rich fertile **brown earth**
- **Mediterranean biome:** 30°–40° latitudes, **western** margins of continents • **dry summer, winter rain from Westerlies** 4×
- **Mediterranean — vegetation:** citrus fruits (orange, lemon) • grapes, olive, fig • cork oak • deep roots, thick bark • 'orchards of the world', wine industry • fire-tolerant scrub — chaparral (California), maquis (Europe), fynbos (S. Africa), mallee (Australia), matorral (Chile)
- **Savanna:** tropical grassland • distinct wet & dry seasons, fires common • **elephant grass** (3–4 m) + scattered trees (acacia, baobab)
- **Savanna — fauna & names:** **'Big Game Country'** • African elephant, giraffe, zebra, lion, cheetah, ostrich, emu • llanos (Venezuela), campos (Brazil)
- **Temperate grasslands:** continental climate • rainfall 25–75 cm • short, soft, nutritious grass • black **chernozem** soil • **'bread basket of the world'** (wheat) • bison, prairie dog, antelope, wild horses
- **Temperate grasslands — regional names: Prairies** (North America) • **Pampas** (Argentina) • **Steppes** (Eurasia) • **Veld** (S. Africa) • **Downs** (Australia) • **Puszta** (Hungary) • **Canterbury** (New Zealand) 6×
- **Tropical rainforest:** 25°–30°C year-round • daily convectional rain, 200 cm+ • no dry season • called **Selvas** in Amazon
- **Rainforest — flora & fauna: stratified (layered)** • hardwoods — mahogany, ebony, rosewood, rubber, cinchona • lianas (woody climbers), epiphytes • monkeys, sloth, toucan, frogs • **terrestrial biome with highest biodiversity** 5×
- **Rainforest — soil: laterite** • leaching makes it acidic, nutrient-poor • rapid decomposition of humus • nutrients stored in tree biomass, not in soil
- **Desert:** rainfall below 25 cm • hot deserts at 15°–30°, western margins of continents • xerophytes — cactus (opuntia), khejri, babul; succulent stems, spine-like leaves

Points to Note

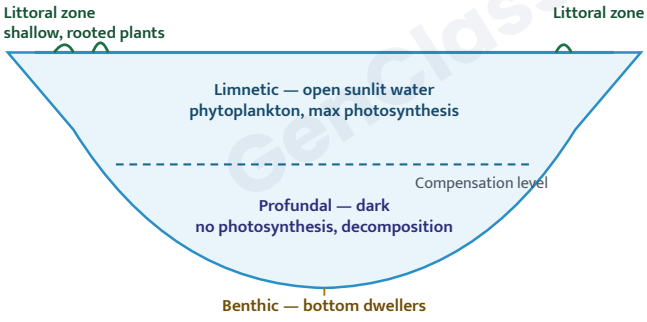
1. Mediterranean rain falls in **winter**, summer is dry — not the reverse
2. Rainforest is the densest, yet its soil is **nutrient-poor** — regrowth after clearing is very difficult
3. Largest = Taiga • highest diversity = rainforest • lowest = tundra
4. Taiga only in the Northern Hemisphere • larch is a conifer yet deciduous
5. Reindeer (Eurasia) and caribou (North America) are the same species

Biome	Soil / Ground	Identity
Tundra	Permafrost	Treeless
Taiga	Podzol	Largest terrestrial
Temperate grassland	Chernozem	Bread basket
Tropical rainforest	Laterite	Highest diversity

Aquatic Ecosystems and Their Zones 5×

- **Lentic (standing water):** lake • pond • swamp • marsh
- **Lotic (flowing water):** river • spring • stream
- **Plankton:** microscopic organisms drifting passively with currents • phytoplankton & zooplankton
- **Nekton:** larger organisms swimming actively against currents — fish, whale, dolphin, turtle
- **Benthos:** bottom-dwellers — crab, starfish, oyster, coral
- **Neuston:** at the air–water interface (surface) — water strider, beetles
- **Periphyton:** attached to aquatic plants/submerged objects — algae, hydra
- **Littoral zone:** shallow near-shore zone • light reaches the bottom • rooted plants
- **Limnetic zone:** open water up to light penetration • plankton-dominated • lower limit = **compensation level** (photosynthesis = respiration)
- **Profundal zone:** deep, lightless zone • only respiration & decomposition • bottom = benthic zone
- **Ocean light zones:** euphotic (photic) up to about 200 m • disphotic (twilight) 200–1,000 m • aphotic below 1,000 m
- **Thermal stratification of a lake:** epilimnion (warm upper) • thermocline (sharp temperature change) • hypolimnion (cold lower)

Zones of a Lake (Lentic)



Photic zone ≈ down to compensation level; ~200 m in oceans

Points to Note

1. Plankton **drift**, nekton **swim** — fish, whale = nekton
2. Neuston = at the **surface** • periphyton = **attached** • benthos = at the **bottom**
3. No photosynthesis in the profundal zone
4. The ~200 m photic depth is for the **sea** • in a lake the limnetic zone extends only to the compensation level

Coral Reefs and Coral Bleaching 8×

- **Nature of coral:** polyp animals of **Cnidaria (Coelenterata)** + dinoflagellate alga **zooxanthellae** • mutualism
- **Roles:** polyp → $CaCO_3$ exoskeleton & shelter • zooxanthellae → food by photosynthesis (about 90% of energy) & colour
- **'Rainforests of the sea':** less than 0.1% of ocean floor • about 25% of marine species
- **Ideal conditions:** 30° N–30° S • water 20°C–25°C (not below 20°C) • depth less than 50–70 m (about 200 ft) • salinity 27–35 ppt
- **Clear, sediment-free water:** silt clogs polyp openings • no corals at river mouths (fresh water + silt)
- **Types of reefs:** fringing • barrier (lagoon in between) • atoll (ring-shaped) • Darwin's subsidence theory
- **India's 4 major reef areas:** **Gulf of Mannar** (Tamil Nadu — fringing) • **Gulf of Kachchh** (Gujarat — fringing) • **Andaman & Nicobar** (fringing + barrier; largest area & diversity) • **Lakshadweep** (entirely atolls) 5×
- **Minor patches & exceptions:** Malvan, Vijaydurg (Maharashtra), Netrani Island (Karnataka) • no corals in **Sundarbans** & **Gulf of Khambhat** — heavy silt + fresh water 4×
- **Coral bleaching:** rising sea temperature (global warming, El Niño), acidification, UV radiation, pollution → zooxanthellae expelled → coral turns **white** → prolonged stress causes death by starvation 4×
- **Fourth global bleaching event:** since 2023 • confirmed by NOAA (April 2024) • most extensive on record

Points to Note

1. Coral = **animal**, not a plant — food & colour come from symbiotic algae
2. Bleaching ≠ instant death — corals can recover if the stress ends
3. Gulf of Kachchh has corals, **Gulf of Khambhat** (Narmada–Tapi silt) does not
4. Lakshadweep = atolls • Andaman & Nicobar = fringing + barrier • Mannar & Kachchh = fringing

Mangroves and Wetlands 7×

- **Mangroves:** halophytes • tidal, saline, oxygen-poor marshy coasts & deltas • salt glands in leaves
- **Vivipary:** seeds germinate while still in the fruit on the parent tree • reason — saline, anoxic swamp below
- **Pneumatophores (breathing roots):** peg-like roots rising above the mud into air • oxygen uptake
- **Stilt & buttress roots:** mechanical support against tidal waves & soft mud
- **Bio-shield:** protection from coastal erosion, cyclones & tsunami 2×
- **Mangrove cover (ISFR 2023):** 4,992 sq km (about 0.15% of country) • **West Bengal** (about 42%) > **Gujarat** (Gulf of Kachchh) > **Andaman & Nicobar** > **Andhra Pradesh** (Godavari–Krishna delta, Coringa)
- **Sundarbans:** world's largest mangrove forest (larger part in Bangladesh) • named after Sundari tree (Heritiera fomes) • Royal Bengal tiger • India's largest Ramsar site
- **Other mangrove sites:** **Bhitarkanika** (Odisha — India's 2nd largest; Brahmani–Baitarani delta; saltwater crocodiles) • **Pichavaram** (Tamil Nadu) • **Chorao Island** (Goa — Salim Ali Bird Sanctuary) • Vembanad (Kerala) • Achara–Ratnagiri (Maharashtra) 4×
- **Ramsar Convention:** 2 Feb 1971, Ramsar (Iran) • World Wetlands Day 2 Feb • India **103** sites (Oct 2026) — Tamil Nadu 20 > Uttar Pradesh 14

Points to Note

1. Bhitarkanika = **2nd** largest mangrove (1st Sundarbans) — not Pichavaram
2. Order: West Bengal > Gujarat > Andaman & Nicobar > Andhra Pradesh
3. Chorao Island = **Goa**, not Maharashtra
4. Pneumatophores = for breathing • stilt/buttress roots = for support

One-line Facts

- **Seagrass:** not algae but true **flowering plants** (angiosperms) • food of the threatened dugong (sea cow) • first Dugong Conservation Reserve — Palk Bay, Tamil Nadu (2022)
- **Kelp forests:** underwater forests of giant brown algae in cold, nutrient-rich shallow seas
- **Great Barrier Reef:** world's largest coral reef • north-east coast of Australia (Queensland, Coral Sea) 3×
- **Shola forests:** Nilgiri, Anamalai, Palani hills • stunted evergreen forests in valleys + montane grasslands on slopes (Southern montane wet temperate forest)
- **Wetlands:** 'kidneys of the landscape' & 'biological supermarkets' — water purification + rich food webs 3×

Part 2: Biodiversity & Conservation

04 Biodiversity: Concept, Measurement and Global Patterns

PYQ 22×

Terminology and Basic Principles

- **Term 'Biological diversity':** **Raymond F. Dasmann** (1968, 'A Different Kind of Country') • scientific usage — Thomas Lovejoy (1980)
- **Term 'Biodiversity':** **Walter G. Rosen** (**1985**) • popularised by entomologist **E. O. Wilson** ('Father of Biodiversity') 4×
- **Genetic diversity:** gene variation within a species • India has 50,000+ rice and 1,000+ mango varieties • Himalayan Rauwolfia vomitoria varies in 'reserpine' content • aids adaptation
- **Species diversity:** number of species (**richness**) + relative abundance (**evenness**) • amphibian diversity higher in Western Ghats than Eastern Ghats
- **Ecological diversity:** variety of ecosystems in a region — desert, rainforest, mangrove, coral reef, wetland, alpine meadow • India > Norway
- **Species on Earth:** a little over 1.5 million described • Robert May's estimate about 7 million • over 70% animals (insects over 70% of these) • plants not more than 22%
- **Rivet popper hypothesis:** **Paul Ehrlich** • ecosystem = aeroplane, species = rivets • removal of key rivets collapses the ecosystem • David Tilman — more species → greater stability & productivity

Points to Note

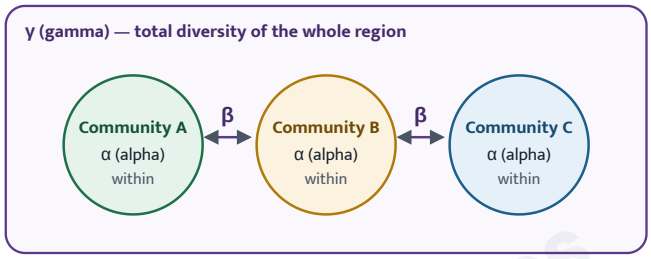
1. 'Biological diversity' = Dasmann • 'Biodiversity' = Rosen • popularised = Wilson
2. Richness = **number** • evenness = balance of **relative abundance**

3. Three levels, small to large: genetic → species → ecological

Measurement of Biodiversity (Whittaker, 1972) 6×

- **Alpha (α):** number of species **within one habitat/community** — local richness
- **Beta (β):** rate of **species turnover between** adjacent habitats or along a gradient • compares unique species of two communities
- **Gamma (γ):** **total** diversity of all habitats in a large region/landscape
- **Diversity indices:** Shannon–Wiener • Simpson — measure both richness & evenness

Three Levels of Diversity: α, β, γ



β (beta) = species turnover between communities · Whittaker, 1972

Points to Note

1. Beta = **difference between** two habitats — greater the difference, higher the beta diversity
2. Whittaker also gave the five-kingdom classification (1969)

Biodiversity Gradients 7×

- **Latitudinal gradient:** diversity **increases** from poles towards the **equator** • highest — **tropical rainforests** (Amazon — 'lungs of the Earth') & **coral reefs**
- **Bird species:** Colombia about 1,400 • New York (41° N) 105 • Greenland (71° N) 56 • India 1,200+
- **Altitudinal gradient:** diversity **decreases** with altitude — falling temperature, harsh weather • about 6.5°C drop per 1,000 m
- **Why tropics are richer:** undisturbed by glaciation → millions of years of uninterrupted speciation • less seasonal, stable environment → niche specialisation • more solar energy & water → high productivity
- **Species–area relationship:** **Alexander von Humboldt** • log S = log C + Z log A • Z usually 0.1–0.2 • 0.6–1.2 for very large areas like continents

Points to Note

1. Towards the equator → diversity **rises** • higher altitude → diversity **falls**
2. Tropics are richer because they were **not affected** by ice ages

Special Ecological Types of Species 5×

- **Endemic:** confined to one geographic region • lion-tailed macaque, Nilgiri tahr (Western Ghats) • Nicobar megapode (Nicobar) • pygmy hog (Assam)
- **Keystone:** term by **Robert Paine (1969)** • low numbers/biomass but **disproportionately large** effect on ecosystem • removal may collapse the ecosystem
- **Keystone examples:** top predators (tiger, lion, sea otter, starfish) • bees & pollinators • fig tree • African elephant
- **Indicator:** highly sensitive to subtle environmental change • early warning
- **Indicator examples:** **lichens** — SO2 pollution (high SO2 → no lichens) • Gangetic dolphin, gharial — clean river water • frogs/amphibians — chemical pollution & UV radiation • trout — high dissolved oxygen • Tubifex worm, E. coli — organic/faecal pollution 5×
- **Umbrella:** large home range • protecting it protects thousands of species • tiger, giant panda
- **Flagship:** charismatic, popular — symbol of a campaign • WWF's **giant panda** • in India tiger, one-horned rhino

Points to Note

1. Keystone = **role** • umbrella = **home range** • flagship = **popularity** — tiger is all three
2. Lichen = air pollution • trout = clean water • Tubifex = polluted water
3. Nicobar megapode is endemic to **Nicobar** — not Andaman

Megadiverse Countries and Biogeographic Zones of India

- **Megadiverse countries:** **Conservation International**, 1998 • **17 countries** • criteria — at least **5,000 endemic plants** + a marine ecosystem within borders
- **India:** **2.4%** of world's land • about **7–8%** of recorded biodiversity (NCERT: 8.1%) • among 17 megadiverse countries • about 45,000 plant species, nearly twice as many animals 4×
- **Biogeographic zones:** Rodgers & Panwar (Wildlife Institute of India, 1988) • **10 zones**, 27 provinces
- **10 zones:** Trans-Himalaya • Himalaya • Indian Desert • Semi-arid • Western Ghats • Deccan Peninsula • Gangetic Plain • North-East • Islands • Coasts
- **Area:** largest **Deccan Peninsula** (about 42%) • second Semi-arid (about 16.6%) • smallest **Islands** (about 0.3%)

Points to Note

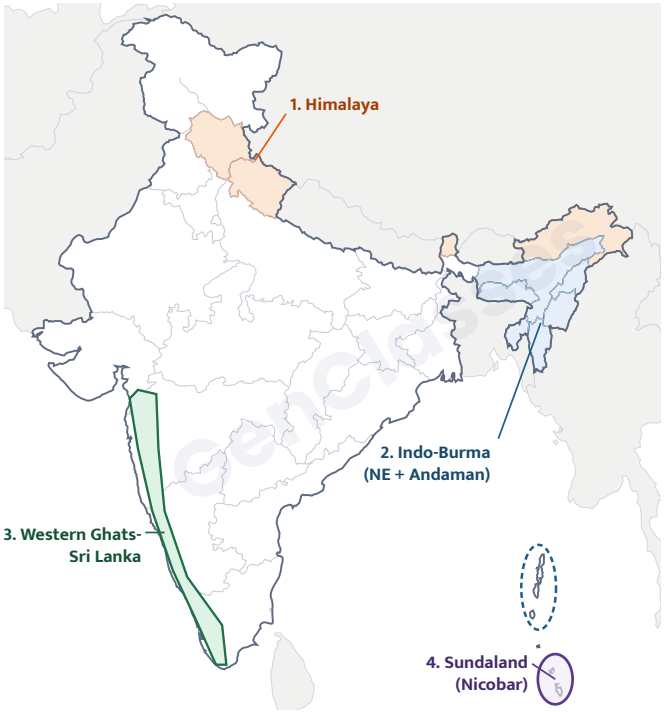
1. Megadiverse countries = Conservation International • hotspots = Norman Myers
2. LMMC (Like-Minded Megadiverse Countries) — Cancun Declaration 2002, India a member; separate from CI's list of 17
3. 2.4% is India's **land area**; its share of biodiversity is 7–8%

Biodiversity Hotspots 14×

- **Proposed by:** British ecologist **Norman Myers**, **1988** • adopted by Conservation International from 1989 4×
- **Criterion 1 — endemism:** at least **1,500 endemic vascular plants** (over 0.5% of world total)
- **Criterion 2 — threat:** at least **70% of original vegetation lost** (30% or less remaining)
- **World:** **36 hotspots** (36th North American Coastal Plain, 2016) • about 2.4% of Earth's land • yet over half of the world's plant species are endemic to them
- **India's 4 hotspots:** Himalaya • Indo-Burma • Western Ghats & Sri Lanka • Sundaland 9×
- **Himalaya:** entire Indian Himalaya (J&K, Ladakh, Himachal, Uttarakhand, Sikkim, Arunachal, Darjeeling) + parts of Pakistan, Nepal, Bhutan, Tibet, Myanmar • snow leopard, musk deer, hangul, red panda, Himalayan monal, western tragopan

- **Indo-Burma:** North-East India south of the Brahmaputra + **Andaman** • Myanmar, Thailand, Vietnam, Laos, Cambodia, southern China • sangai deer (Manipur), **hoolock gibbon** (India's only ape), white-winged wood duck
- **Western Ghats & Sri Lanka:** Gujarat–Maharashtra border (Tapi) to Kanyakumari — Sahyadri, Nilgiri, Anamalai, Cardamom + Sri Lanka • richest diversity in the peninsula • lion-tailed macaque, Nilgiri tahr, Malabar civet, great Indian hornbill, purple frog
- **Sundaland:** only India's **Nicobar** + Indonesia, Malaysia, Singapore, Brunei, southern Thailand • Nicobar megapode, Nicobar crab-eating macaque, leatherback turtle, orangutan
- **Old classification:** earlier 'Eastern Himalaya' & 'Western Ghats' • now entire Himalaya = Himalaya, North-East = Indo-Burma 6×

India's 4 Biodiversity Hotspots



World: 36 (Conservation International) · criteria: ≥1,500 endemic vascular plants + ≥70% original habitat lost · concept: Norman Myers (1988) · shading indicative

Points to Note

1. If options give only 'Eastern Himalaya' & 'Western Ghats', choose them — UPPSC has accepted this many times
2. **Andaman** = **Indo-Burma** • **Nicobar** = **Sundaland**
3. Eastern Ghats, Aravalli, Thar, Sundarbans — **not** hotspots
4. Criterion is 70% **loss** (30% remaining) — not 30% loss
5. The UN does not declare hotspots • 2013 status of Great Nicobar = UNESCO **Biosphere Reserve**

Hope Spots (Marine)

- **Proposed by:** marine biologist **Sylvia Earle** — Mission Blue (2009), with IUCN
- **Meaning:** ocean areas critical to marine ecosystem health, meriting special protection
- **India's 2 Hope Spots:** **Andaman & Nicobar** • **Lakshadweep**

Points to Note

1. Hope Spot = **marine** area • hotspot = mainly terrestrial, based on endemic plants
2. Hope Spot — Sylvia Earle • hotspot — Norman Myers • megadiverse countries — Conservation International

One-line Facts

- **International Day for Biological Diversity:** **22 May** — CBD text adopted at Nairobi on 22 May 1992 4×
- **UN Decades:** 2011–2020 Decade on Biodiversity • 2021–2030 **Decade on Ecosystem Restoration**
- **Silent Valley:** Kerala, **Palakkad**, Nilgiri hills • tropical evergreen forest • lion-tailed macaque • **Kunthipuzha river** • National Park 1984 5×
- **Valley of Flowers:** Uttarakhand, **Chamoli** • alpine flowers, musk deer • UNESCO World Heritage Site with Nanda Devi
- **Most species-rich groups in India:** insects (Arthropoda) • among plants angiosperms • fungi also very rich

05 Biodiversity Loss, IUCN & Conservation Methods

PYQ 25×

Causes of Biodiversity Loss — the 'Evil Quartet'

- **Evil Quartet:** 4 major human-caused drivers of extinction • term — **Jared Diamond**
- **Habitat loss & fragmentation:** the **biggest** cause • tropical rainforests shrank from 14% to 6% of land surface 6×
- **Over-exploitation:** greed-driven hunting and over-harvesting • extinct — **Steller's sea cow** • **Passenger pigeon**
- **Alien species invasion:** species brought in from outside • spreads fast for lack of natural enemies • native species wiped out
- **Invasive plants (India):** Water hyacinth (Eichhornia crassipes — **'Terror of Bengal'**) • Carrot grass (Parthenium hysterophorus) • Lantana camara • Vilayati babool (Prosopis juliflora) • Senna spectabilis (threat in Nilgiri Biosphere Reserve)
- **Invasive animals:** African catfish (Clarias gariepinus — threat to native catfishes of Indian rivers) • Nile perch (Lake Victoria, Africa — 200+ cichlid fishes extinct)
- **Co-extinction:** when one species dies out, the species **obligately** dependent on it also dies out • host fish → specific parasites • plant ↔ specific pollinator • Dodo (Mauritius) → Calvaria tree stopped germinating
- **Other drivers:** Climate change • Pollution • IPBES (2019) ranking — Land/sea-use change > Direct exploitation > Climate change > Pollution > Invasive species
- **Sixth mass extinction:** rate **100–1000 times** the pre-human rate • recent extinctions — Dodo (Mauritius) • Quagga (Africa) • Thylacine (Australia) • Steller's sea cow (Russia) • Bali, Javan, Caspian tiger subspecies

Points to Note

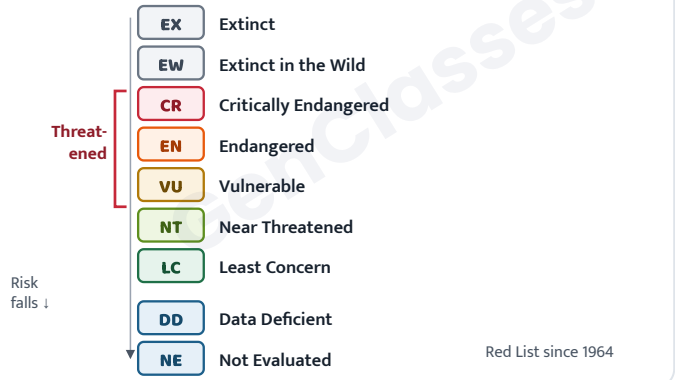
1. Biggest cause = habitat loss & fragmentation — not over-exploitation or invasive species
2. 'Evil Quartet' — Jared Diamond • 'HIPPO' (Habitat, Invasive species, Pollution, Population, Over-harvesting) — E.O. Wilson
3. 'Terror of Bengal' = water hyacinth, not carrot grass • Nile perch = Lake Victoria, not India
4. Co-extinction needs obligate dependence — parasite-host, specific pollinator-plant

International Union for Conservation of Nature (IUCN) & Red Data Book 8×

- **Founded:** **5 October 1948** • Fontainebleau (France) • original name IUPN → IUCN in 1956
- **Headquarters:** **Gland (Switzerland)**

- **Nature: not** a UN body • members = governments + NGOs • world's first global environmental organisation, largest environmental network • UN General Assembly observer (1999)
- **Red List:** since 1964 • global list of species' extinction risk • Red Data Book = its book form
- **Red Data Book pages: Pink** — critically endangered • **Green** — species that have recovered
- **India's Red Data Book:** plants — Botanical Survey of India (BSI) • animals — Zoological Survey of India (ZSI)
- **9 categories (in order):** Extinct (EX) • Extinct in the Wild (EW) • **Critically Endangered (CR)** • **Endangered (EN)** • **Vulnerable (VU)** • Near Threatened (NT) • Least Concern (LC) • Data Deficient (DD) • Not Evaluated (NE)
- **Threatened:** only **CR + EN + VU**
- **Extinct (EX):** no reasonable doubt the last individual has died • not found even by exhaustive surveys across the historic range
- **Extinct in the Wild (EW):** survives only in captivity/cultivation or as a naturalised population outside its past range
- **Number criterion (mature individuals):** CR < 50 • EN < 250 • VU < 1,000

IUCN Red List Categories



Points to Note

1. IUCN is not a UN agency — only an observer • HQ Gland, not Geneva
2. 'Threatened' = only CR, EN, VU — NT and DD are not included
3. Pink pages = critically endangered • Green pages = recovered species — not the reverse
4. No fixed time limit for declaring extinction — 'not seen for 50 years' is the old criterion
5. EW ≠ EX — an EW species is still alive in captivity/cultivation

Major Critically Endangered (CR), Endangered (EN) & Vulnerable (VU) Species of India 5×

- **CR — Mammals: Hangul/Kashmir stag** (mainly Dachigam National Park • state animal of J&K) • Andaman shrew • Kondana rat (Sinhagad plateau, Pune) • Malabar large-spotted civet
- **CR — Birds: Great Indian Bustard (Godawan)** — state bird of Rajasthan, Desert National Park • Bengal florican • Jerdon's courser (Sri Lankamalleswara Wildlife Sanctuary, Andhra Pradesh) • Siberian crane • Spoon-billed sandpiper
- **CR — Vultures:** White-rumped • Indian • Slender-billed • Red-headed • cause — veterinary drug **diclofenac** (veterinary use banned 2006; alternative meloxicam)
- **CR — Reptiles: Gharial (Gavialis gangeticus)** — Chambal (largest) • Girwa (Katarniaghat, UP) • Son • Gandak • Ramganga • Hawksbill turtle • Red-crowned roofed turtle (Chambal)
- **EN — Mammals:** Royal Bengal tiger • Asiatic lion (only Gujarat — Gir landscape; 891, 2025) • **Gangetic dolphin (Susu)** • Red panda • **Sangai deer** (Keibul Lamjao — floating National Park, Loktak, Manipur) • Nilgiri tahr • Lion-tailed macaque • Asian elephant • Golden langur • **Pygmy hog** (Manas, Assam)
- **EN — Others:** Dhole (wild dog) • Indian pangolin • Forest owlet
- **VU: One-horned rhinoceros** • Snow leopard • Cheetah (species) • Dugong (sea cow) • **Olive Ridley turtle** 4×
- **Olive Ridley — 'Arribada' (mass nesting):** Odisha — Gahirmatha • Rushikulya • Devi river mouth
- **VU — Others:** Gaur • Sloth bear • Leopard • Barasingha (state animal of UP) • Clouded leopard

Points to Note

1. One-horned rhino = VU, not EN • Snow leopard VU (since 2017) • Pygmy hog now EN (since 2019, earlier CR)
2. Tiger, Asiatic lion, elephant, Gangetic dolphin = EN • Great Indian Bustard, gharial, hangul, vultures = CR
3. Blackbuck now LC — not on the threatened list
4. Gangetic dolphin = National Aquatic Animal (2009) • State Aquatic Animal of UP (2023)

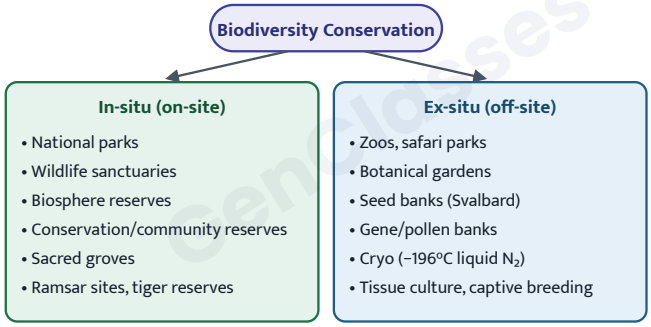
Conservation Methods: In-situ vs Ex-situ 13×

Basis	In-situ	Ex-situ
Location	Within the natural habitat	Outside habitat, human-controlled
What is saved	Whole ecosystem, food web, evolutionary processes	Selected species/genes
Cost	Cheap, more effective	Expensive
When	Routine, long-term measure	Species on the brink — life-saving

- **In-situ:** National Park • Wildlife Sanctuary • Biosphere Reserve • Sacred groves and sacred lakes • Community & Conservation Reserves • Marine Protected Areas • Ramsar wetlands • Biodiversity hotspots
- **Ex-situ:** Botanical garden • Zoo and safari park • Seed bank • Gene bank • Cryopreservation • Tissue culture • Artificial insemination • Aquarium • Arboretum • Nursery
- **Sacred groves:** Meghalaya — Khasi & Jaintia hills • Rajasthan — Aravalli (Oran) • Karnataka & Maharashtra — Western Ghats (Devrai) • Chhattisgarh — Sarguja, Bastar • Kerala — Kavu
- **Cryopreservation:** gametes (pollen, sperm, ova), embryos, tissues • kept alive for years in **liquid nitrogen** at **−196°C** 4×
- **National Seed Vault: Chang La (Ladakh)** • DRDO + NBPGR, 2010 • world's largest — **Svalbard (Norway)**
- **National Gene Bank:** NBPGR, New Delhi
- **National Park (107):** conserves the whole ecosystem, not focused on one species • grazing, wood collection, private rights **completely prohibited** • boundaries rigid, fixed by legislative process
- **Wildlife Sanctuary (570+):** usually for a particular species/group • limited grazing and activity with Chief Wildlife Warden's permission • boundaries less rigid
- **Biosphere Reserve (18 · 13 in UNESCO WNBR):** integrated development of biodiversity + landscape + local/tribal communities • UNESCO **MAB (1971)** norms

- **3 zones of a Biosphere Reserve:** Core — no human activity • Buffer — research, education, tourism • Transition — settlements, farming, economic activity

In-situ vs Ex-situ Conservation



In natural habitat = in-situ · outside habitat = ex-situ

Points to Note

1. Ramsar wetlands, sacred groves, hotspots, Marine Protected Areas = in-situ
2. Seed/gene banks, cryopreservation, tissue culture, zoos, botanical gardens = ex-situ
3. Cryopreservation = liquid nitrogen, −196°C — not dry ice
4. National Parks and Sanctuaries — under the Wildlife (Protection) Act 1972 • Biosphere Reserves have no provision in it (UNESCO MAB, designated by the Centre)

One-line Facts

- **Indian Rhino Vision (IRV 2020):** launched 2005 • target of 3,000 one-horned rhinos across 7 protected areas of Assam
- **National Rhino Conservation Strategy:** 2019 • for the one-horned rhinoceros

06 Protected Areas and Species Conservation Projects of India

PYQ 20×

Protected Area Network

- **Legal basis:** Wildlife (Protection) Act, 1972 • 4 categories — National Park • Wildlife Sanctuary • Conservation Reserve • Community Reserve (last two added by 2002 amendment)
- **Total (March 2026): 1,034** protected areas • National Parks **107** (107th Similipal, Odisha, 2025; 103 in 2014)
- **National Park:** protects an entire habitat • human activity, grazing, wood collection, private land rights **completely prohibited**
- **Wildlife Sanctuary:** often focused on a particular species • regulated grazing and some human activities allowed with Chief Wildlife Warden's permission
- **Status change:** Sanctuary → National Park (upgradation) possible • National Park → Sanctuary (downgradation) not possible
- **Boundary change:** only on recommendation of **National Board for Wildlife (NBWL)** • NBWL chairperson — Prime Minister
- **Conservation Reserve:** on government land, after consulting local communities • often a corridor between parks and sanctuaries
- **Community Reserve:** on private/community land, voluntarily offered by the community
- **Eco-Sensitive Zone (ESZ):** generally up to 10 km around protected areas • notified under Environment (Protection) Act, 1986 • acts as a "shock absorber"

Comparing Protected Areas (Wildlife Act)

	National Park	Wildlife Sanctuary	Conservation Reserve	Community Reserve
Land	Govt	Govt	Govt	Private/community
Grazing/rights	Fully barred	With permit	Rights retained	Community-run
Introduced	1972	1972	2002 amend.	2002 amend.
Status	No downgrade	Can be upgraded	Corridor/buffer	Voluntary

Boundary change only on NBWL recommendation · Biosphere reserves are not a category under this Act

Points to Note

1. National Park — grazing/private rights never; Sanctuary — allowed with Warden's permission
2. A Sanctuary can become a National Park, never the reverse — boundaries cannot change without NBWL recommendation
3. Biosphere Reserve is not a category under the Wildlife Act — declared under a Central scheme (UNESCO MAB basis)
4. ESZ comes from Environment (Protection) Act, 1986, not from the Wildlife Act

National Parks: First, Largest and Special 20×

- **First National Park: Jim Corbett (Uttarakhand)** • 1936 as 'Hailey National Park' (after Governor Malcolm Hailey) • 1954-55 'Ramganga National Park' • 'Jim Corbett' from 1955-56 • **Ramganga river** flows through it 5×
- **Largest: Hemis (Ladakh)** • about 4,400 sq km • snow leopard capital
- **Smallest: South Button Island (Andaman & Nicobar)** • only 0.03 sq km
- **World's only floating park: Keibul Lamjao (Manipur)** • part of **Loktak Lake** • floating biomass-vegetation = 'Phumdi' • only natural habitat of Manipur's state animal Sangai deer (Dancing Deer) 5×
- **First Marine National Park: Gulf of Kachchh (Jamnagar, Gujarat — 1982)** 3×
- **Other Marine National Parks:** Gulf of Mannar (Tamil Nadu) • Mahatma Gandhi Marine (Wandoor, Andaman) • Rani Jhansi Marine (Andaman)
- **Marine Sanctuaries (not parks):** Gahirmatha (Odisha) • Malvan (Maharashtra)
- **Most National Parks:** 1. **Madhya Pradesh (11)** • 2. Andaman & Nicobar (9) • 3. Assam (7 — 6th Raimona, 7th Dehing Patkai)
- **Natural World Heritage parks:** Kaziranga, Manas, Keoladeo (1985) • Sundarbans (1987) • Nanda Devi-Valley of Flowers (1988/2005) • Great Himalayan (2014) • Khangchendzonga (2016, **mixed**)

Points to Note

1. Largest National Park Hemis ≠ largest Tiger Reserve Nagarjunsagar-Srisailam ≠ largest Biosphere Reserve Rann of Kachchh
2. Gahirmatha and Malvan are Marine Sanctuaries, not National Parks
3. Most parks: among states Madhya Pradesh, among UTs Andaman & Nicobar; Assam third
4. Phumdi = floating vegetation of Loktak • Sangai (Keibul Lamjao) ≠ Hangul (Dachigam)

National Parks: State-wise Matching 10×

- **Jammu & Kashmir:** **Dachigam** (only habitat of Hangul/Kashmir stag) • Kishtwar • Salim Ali (City Forest, Srinagar) 4×
- **Himachal Pradesh:** **Great Himalayan** (UNESCO World Heritage) • **Pin Valley** (cold desert) • Inderkilla • Khirganga • Simbalbara
- **Uttarakhand:** Jim Corbett • **Rajaji** (Asian elephant) • Nanda Devi • Valley of Flowers • Gangotri • Govind Pashu Vihar
- **Assam:** **Kaziranga** (about two-thirds of world's one-horned rhinos; Brahmaputra bank) • **Manas** (tiger, pygmy hog, golden langur) • Orang (mini Kaziranga) • Nameri • Dibru-Saikhowa (feral horses) • Raimona (golden langur) • Dehing Patkai (tropical rainforest)
- **Arunachal Pradesh:** **Namdapha** (Eastern Himalaya; tiger, leopard, snow leopard, clouded leopard — world's only park with all 4 big cats together; altitude about 200 to 4,500 m; flying squirrel) • Mouling 3×
- **Rajasthan:** **Keoladeo Ghana, Bharatpur** (formerly Bharatpur Bird Sanctuary; Banganga-Gambhir confluence; wintering of Siberian crane, none since 2002; Montreux Record) • Ranthambore • Sariska • Mukundra Hills (Darrah) • **Desert NP** (Great Indian Bustard/Godawan)
- **Gujarat:** **Gir** (only natural home of Asiatic lion) • **Blackbuck, Velavadar** (blackbuck) • Vansda • Marine (Gulf of Kachchh) • Wild Ass Sanctuary = Little Rann of Kachchh
- **Madhya Pradesh:** **Kanha** (Barasingha/swamp deer; first park with official mascot 'Bhoorsingh the Barasingha') • Bandhavgarh (white tigers) • Panna (Ken river flows through) • Pench (Priyadarshini) • Satpura • Madhav • Sanjay • Van Vihar • Fossil (Mandla) • Dinosaur Fossil (Dhar) • **Kuno** (cheetah)
- **Chhattisgarh:** Indravati (Kutru) • Kanger Ghati • Guru Ghasidas (Sanjay)
- **Odisha:** Simlipal • **Bhitarkanika** (saltwater crocodile; Olive Ridley turtles on nearby Gahirmatha coast)
- **West Bengal:** **Sundarbans** (mangrove-dwelling, expert-swimmer Royal Bengal tiger) • Buxa • Gorumara • Neora Valley • **Singalila** (red panda) • Jaldapara (one-horned rhino)
- **Karnataka:** Bandipur • Nagarhole (Rajiv Gandhi) • **Bannerghatta** (India's first butterfly park) • Anshi (Kali) • Kudremukh
- **Kerala:** Periyar (elephant-tiger; Periyar river) • Silent Valley • **Eravikulam** (largest Nilgiri tahr population; **Anamudi**, highest peak of South India) • Mathikettan Shola • Anamudi Shola • Pambadum Shola

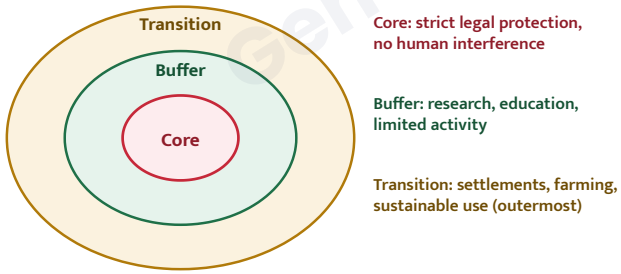
Points to Note

1. Hangul — Dachigam • Sangai — Keibul Lamjao • Barasingha — Kanha • Nilgiri tahr — Eravikulam • Great Indian Bustard — Desert NP
2. Red panda — Singalila/Neora Valley (West Bengal) • Golden langur — Manas/Raimona (Assam)
3. Rhino not only in Assam — also Jaldapara and Gorumara (West Bengal)
4. Ken in Panna, Ramganga in Corbett, Periyar river in Periyar
5. Wild ass in Little Rann of Kachchh; Great Rann of Kachchh = largest Biosphere Reserve

Biosphere Reserves: MAB and Zonation 11×

- **Basis:** UNESCO's **'Man and the Biosphere' (MAB) Programme — 1971** • India's National Biosphere Reserve Programme **1986**
- **Declaration:** Central Government (Environment Ministry) • nomination to UNESCO World Network (WNBR) also by the Centre
- **3 functions:** conservation • sustainable development • logistic support (research, monitoring, education)
- **3 concentric zones:** core • buffer • transition 3×
- **Core zone:** innermost, fully protected • **no** human interference except research • usually a National Park/Sanctuary forms it
- **Buffer zone:** surrounds the core • regulated research, environmental education, training, limited tourism and grazing
- **Transition zone:** outermost open area • tribal settlements, agriculture, forestry, sustainable economic activities — in harmony with conservation

Zonation of a Biosphere Reserve



UNESCO MAB Programme (1971) · notified by the Central Govt in India

Points to Note

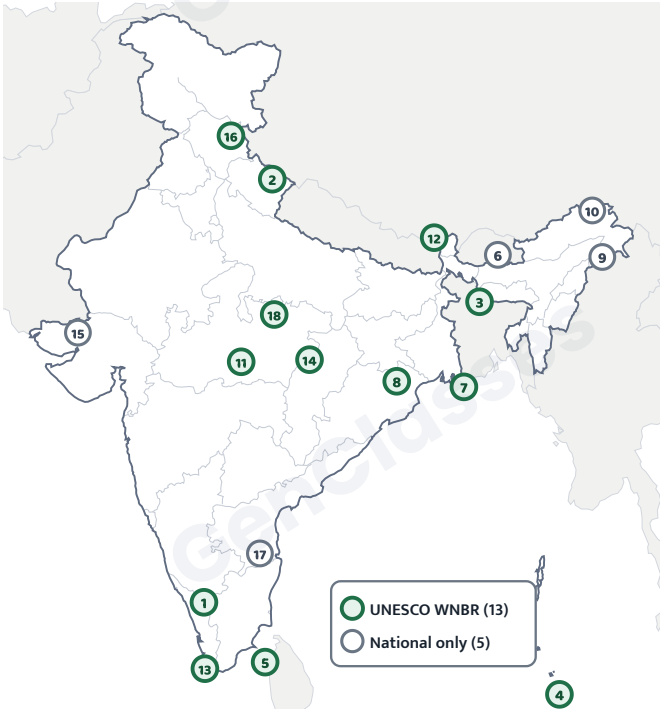
1. Core — only research • tourism-grazing in buffer • settlements-farming in transition
2. Transition zone has no rigid boundary; core and buffer are delimited
3. MAB 1971 (UNESCO), but India's programme 1986 — first BR Nilgiri also 1986
4. Human settlement is legal in a Biosphere Reserve (transition zone), not in a National Park

India's 18 Biosphere Reserves 17×

- **Total:** 18 • **13** in UNESCO World Network (WNBR) (13th — Cold Desert, 27 Sep 2025) • 5 outside
- **Nilgiri (1986):** Tamil Nadu-Kerala-Karnataka • **India's first** BR • first in UNESCO too (2000) • includes Mudumalai, Wayanad, Bandipur, Nagarhole, Silent Valley 5×
- **Nanda Devi (1988):** Uttarakhand (Chamoli, Pithoragarh, Bageshwar) • UNESCO 2004 • snow leopard, musk deer
- **Nokrek (1988):** Meghalaya (Garo Hills) • UNESCO 2009 • red panda • Citrus indica (wild orange) 4×
- **Gulf of Mannar (1989):** Tamil Nadu • UNESCO 2001 • **Dugong** (sea cow) • among the large marine reserves
- **Sundarban (1989):** West Bengal • UNESCO 2001 • Ganga-Brahmaputra delta • Sundari tree • Royal Bengal tiger
- **Great Nicobar (1989):** Andaman & Nicobar • UNESCO 2013 • **southernmost** BR • saltwater crocodile
- **Simlipal (1994):** Odisha (Mayurbhanj) • UNESCO 2009 • melanistic (black) tigers • sal forests
- **Pachmarhi (1999):** Madhya Pradesh (Satpura range) • UNESCO 2009 • includes Bori Sanctuary, Satpura NP • giant squirrel, flying squirrel
- **Khangchendzonga (2000):** Sikkim • UNESCO 2018 • Khangchendzonga NP = India's only **mixed** World Heritage Site (2016)
- **Agasthyamala (2001):** Kerala-Tamil Nadu • UNESCO 2016 • southern tip of Western Ghats • **Kani tribe** • Nilgiri tahr 3×
- **Achanakmar-Amarkantak (2005):** Madhya Pradesh-Chhattisgarh • UNESCO 2012 • source of Narmada, Son, Johila
- **Panna (2011):** Madhya Pradesh • **18th (latest)** BR • UNESCO 2020 (12th) • Ken river flows through 2×

- **Cold Desert (2009):** Himachal (Lahaul-Spiti; Pin Valley NP, Kibber Sanctuary, Chandratal) • UNESCO **2025 (13th)** • snow leopard
- **Manas (1989):** Assam • outside UNESCO
- **Dibru-Saikhowa (1997):** Assam • **smallest** BR (765 sq km) • outside UNESCO
- **Dehang-Debang (1998):** Arunachal Pradesh • outside UNESCO
- **Rann of Kachchh (2008):** Gujarat • **largest** BR (12,454 sq km) • wild ass • outside UNESCO
- **Seshachalam Hills (2010):** Andhra Pradesh • Eastern Ghats • **red sanders** • slender loris • outside UNESCO

India's 18 Biosphere Reserves



1 Nilgiri · 2 Nanda Devi · 3 Nokrek · 4 Great Nicobar · 5 Gulf of Mannar · 6 Manas · 7 Sundarban · 8 Simlipal · 9 Dibru-Saikhowa · 10 Dehang-Debang · 11 Pachmarhi · 12 Khangchendzonga · 13 Agasthyamala · 14 Achanakmar-Amarkantak · 15 Kachchh · 16 Cold Desert · 17 Seshachalam · 18 Panna

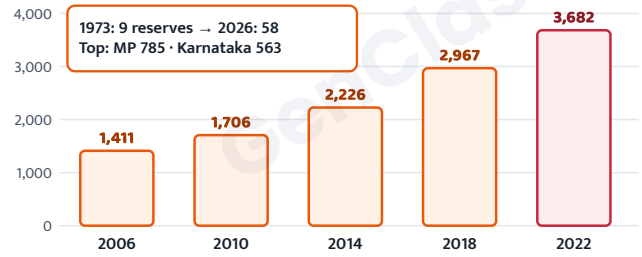
Points to Note

1. UNESCO now 13 — Cold Desert (2025) added; only 5 outside (Assam 2, Arunachal 1, Gujarat 1, Andhra 1)
2. Largest (Kachchh) and smallest (Dibru-Saikhowa) — both outside UNESCO
3. Latest BR Panna (2011) ≠ latest UNESCO entry Cold Desert (2025)
4. Khangchendzonga is India's only mixed World Heritage Site — not the world's only
5. Only Nilgiri spans 3 states; Achanakmar-Amarkantak and Agasthyamala span two

Tiger Reserves and Tiger Estimation

- **NTCA:** on Tiger Task Force (2005) recommendation • statutory body from **2006** via Wildlife Act amendment • chairperson — Union Environment Minister
- **Notification:** by State Government on NTCA advice (Section 38V) • core (critical tiger habitat) + buffer
- **Total Tiger Reserves: 58** • most in Madhya Pradesh (9)
- **Latest (53-55):** 53rd Ranipur (Chitrakoot, UP, 2022) • 54th Veerangana Durgavati (Sagar-Damoh-Narsinghpur, MP, 2023) • 55th Dholpur-Karauli (Rajasthan, 2023)
- **Latest (56-58):** 56th Guru Ghasidas-Tamor Pingla (Chhattisgarh, Nov 2024) • 57th Ratapani (Raisen, MP, Dec 2024) • **58th Madhav (Shivpuri, MP, March 2025)**
- **Largest:** **Nagarjunsagar-Srisailam (Andhra Pradesh)** • then Manas (Assam) • then Guru Ghasidas-Tamor Pingla (Chhattisgarh) 4×
- **Smallest:** **Bor (Maharashtra)** — standard exam answer (smallest when notified, 2014) • by present area Orang (core 79 sq km) & Nameri in Assam are smaller
- **Tiger estimation:** every **4 years** • **M-STrIPES** app (patrolling and ecological monitoring)
- **Tiger numbers:** 2006: 1,411 • 2010: 1,706 • 2014: 2,226 • 2018: 2,967 • **2022: 3,682** (average; minimum 3,167)
- **Estimation 2022 (5th cycle, report 2023):** about **75%** of world's wild tigers in India
- **Most tigers — state:** 1. **Madhya Pradesh (785, 'Tiger State')** • 2. Karnataka (563) • 3. Uttarakhand (560) • 4. Maharashtra (444)
- **Most tigers — reserve:** **Jim Corbett (260)** • Bandipur (150) • Nagarhole (141)
- **International Tiger Day:** **29 July** • from 2010 St. Petersburg Tiger Summit (Russia) • **TX2** goal (double tigers by 2022) — India achieved it in 2018 itself

Tiger Numbers in India (AITE)



2022 (5th cycle): mean 3,682 · minimum 3,167 · ~75% of world's wild tigers

Points to Note

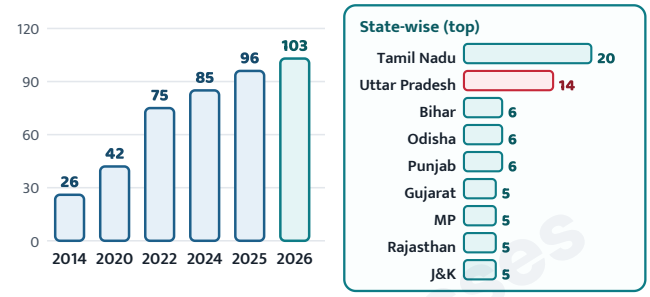
1. Most tigers: state Madhya Pradesh, reserve Jim Corbett; most Tiger Reserves also Madhya Pradesh (57th and 58th both MP)
2. M-STrIPES = software app for patrolling and monitoring tiger reserves, not a scheme
3. NTCA chaired by Environment Minister; NBWL chaired by Prime Minister
4. Tiger Day 29 July ≠ World Elephant Day 12 August ≠ National Dolphin Day 5 October

Ramsar Wetlands and Mangroves 15×

- **Ramsar Convention:** **2 February 1971**, Ramsar (Iran) • in force 21 December 1975 • India party **1 February 1982** • World Wetlands Day 2 February 6×
- **India total:** **103 sites** (15 October 2026) • about 13.88 lakh ha • **3rd** in world (UK 176, Mexico 144) • 1st in Asia
- **State-wise:** **Tamil Nadu 20** (most) • **Uttar Pradesh 14** (2nd) • Bihar, Odisha, Punjab 6 each
- **No site:** Telangana • Meghalaya • Nagaland • Delhi • Puducherry • Lakshadweep • Andaman & Nicobar • DNH-DD
- **First (1 October 1981):** **Chilika Lake (Odisha)** • **Keoladeo National Park (Rajasthan)**
- **Largest:** **Sundarban Wetland (West Bengal)** 4,23,000 ha • 2. Vembanad-Kol (Kerala) • 3. Chilika

- **Smallest: Vembannur Wetland Complex (Tamil Nadu)** 19.75 ha • then Renuka (Himachal) 20 ha
- **Highest altitude:** Tso Kar (Ladakh)
- **Milestones (97-100):** 97th-98th Chhari Dhand (Kachchh) and Patna Bird Sanctuary (Etah) — January 2026 • 99th Shekha Jheel (Aligarh, April 2026) • **100th** Jai Prakash Narayan Bird Sanctuary/Surha Tal (Ballia, 5 June 2026)
- **Milestones (101-103):** 101st Glaw Lake (Lohit, Arunachal's first, 3 August 2026) • 102nd Sukhna Wetland (Chandigarh's first) • **103rd Sauj Jheel (Mainpuri, UP)**
- **Key additions 2024-25:** Ankasamudra, Aghanashini estuary, Magadi Kere (Karnataka) • Karaivetti, Longwood Shola, Nanjarayan, Kazhuveli, Sakkarakottai, Therthangal (Tamil Nadu) • Nagi and Nakti (Jamui, Bihar) • Tawa Reservoir (MP) • **Khecheopalri** (Sikkim's first) • **Udhwa Lake** (Jharkhand's first)
- **Montreux Record:** register of Ramsar sites with serious adverse change in ecological character • India's 2 — **Keoladeo (1990), Loktak (1993)** • Chilika removed 2002 (first in Asia) 4×
- **Wetland City Accreditation (January 2025):** Indore • Udaipur — India's first
- **Ramsar COP:** COP15 Victoria Falls (Zimbabwe, July 2025) • COP16 Panama (2028)
- **Amrit Dharohar (2023):** scheme for conservation and eco-tourism of Ramsar sites
- **Mangroves (ISFR 2023):** total 4,992 sq km • Sundarbans — world's largest mangrove forest (India-Bangladesh) • Bhitarkanika — India's second largest mangrove ecosystem
- **Mangrove initiatives:** MISHTI (Budget 2023-24) • Mangrove Alliance for Climate (COP27, 2022 — India a member) • International Day for Conservation of Mangrove Ecosystem 26 July

India's Ramsar Sites: Growth & State-wise



103 by 15 Oct 2026 · 77 added since 2014 · 1st in Asia, 3rd in world

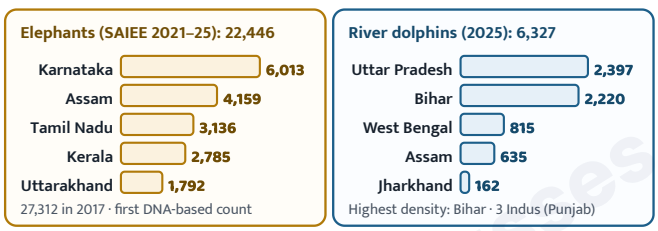
Points to Note

1. Smallest Ramsar site is now Vembannur (19.75 ha); Renuka (20 ha) second
2. Montreux Record has only Keoladeo and Loktak — Chilika out in 2002; Keoladeo is both a first Ramsar site and on Montreux
3. Convention 1971 • in force 1975 • India 1982 — first two sites designated with 1981 date
4. 100th Surha Tal (Ballia) ≠ Sur Sarovar/Keetham (Agra)

Species Conservation Projects (Chronology) 19×

- **Project Hangul (1970):** Kashmir stag (state animal of J&K) • **Dachigam** • J&K Government + IUCN/WWF
- **Gir Lion Project (1972):** Asiatic lion • Gir forest (Junagadh, Gujarat) • 16th estimation (May 2025): **891** (2020: 674)
- **Project Tiger (1 April 1973):** from **Jim Corbett** • 9 reserves initially • first director **Kailash Sankhala** ('Tiger Man of India') • 50 years in 2023 8×
- **Crocodile Conservation Project (1975):** UNDP + FAO assistance • Odisha — Tikarpada (Satkosia Gorge, Mahanadi) and Bhitarkanika • all three species in India — gharial, mugger, saltwater crocodile • key centres National Chambal Sanctuary (MP-Rajasthan-UP), Kukrail (Lucknow)
- **Manipur Thamin/Sangai Project (1977):** brow-antlered deer (Dancing Deer; Manipur state animal) • Keibul Lamjao (south-eastern part of Loktak Lake)
- **Project Rhino (1987):** one-horned rhino • Kaziranga • Pobitora (**highest density**) • Manas (Assam) • Jaldapara (West Bengal)
- **Project Elephant (February 1992):** centrally sponsored • on 1988 Task Force recommendation • first Elephant Reserve Singhbhum (Jharkhand, 2001) • Elephant Reserves **33** (33rd Terai, UP, 2022) 6×
- **Elephant — special: National Heritage Animal (October 2010)** • MIKE (monitoring illegal killing; CITES, in India since 2003; 10 sites, e.g. Shivalik) • Gaj Utsav 2023 (Kaziranga, 30 years of the project) • World Elephant Day 12 August
- **Elephant estimation (SAIEE 2021-25, DNA-based, October 2025): 22,446** (2017: 27,312) • Karnataka 6,013 > Assam 4,159 > Tamil Nadu 3,136 > Kerala 2,785 > Uttarakhand 1,792 • Western Ghats 11,934
- **Red Panda Project (1996):** Padmaja Naidu Himalayan Zoological Park (Darjeeling) + WWF • local name **'cat-bear'** • habitat Sikkim (state animal), Arunachal, Darjeeling
- **Sea Turtle Project (1999):** Environment Ministry + UNDP • nodal WII • Olive Ridley mass nesting (**Arribada**) — Gahirmatha Marine Sanctuary (near Bhitarkanika) and Rushikulya estuary (Odisha)
- **Vulture conservation (2006): diclofenac** banned in veterinary use (2006) • first Action Plan 2006 • current Action Plan **2020-2025** • first vulture conservation-breeding centre Pinjore (Haryana) • world's first Jatayu (red-headed/king vulture) conservation-breeding centre Maharajganj (Farenda, UP) 3×
- **Project Snow Leopard (2009):** 5 Himalayan states (J&K — now incl. Ladakh, Himachal, Uttarakhand, Sikkim, Arunachal) • 2013 **Bishkek Declaration** and GSLEP (12 countries) • first snow leopard conservation centre Uttarkashi (Uttarakhand)
- **Snow leopard estimation (SPAI, January 2024): 718** • Ladakh 477 (highest) • Uttarakhand 124
- **Project Dolphin (15 August 2020):** both river and marine dolphins • Gangetic dolphin (Platanista gangetica, 'Susu') **National Aquatic Animal — 5 October 2009** → 5 October National Dolphin Day 5×
- **Gangetic dolphin — special:** practically blind • hunts by **echolocation** • Vikramshila Gangetic Dolphin Sanctuary (Bhagalpur, Bihar) — India's only dolphin sanctuary • National Dolphin Research Centre (NDRC) Patna
- **Dolphin estimation (first, 3 March 2025): 6,327** (6,324 Gangetic + 3 Indus — Beas, Punjab) • UP 2,397 > Bihar 2,220 > West Bengal 815 > Assam 635 > Jharkhand 162 > MP-Rajasthan 95 > Punjab 3 • highest density in Bihar

Elephant & Dolphin Counts: Top States



- **Project Cheetah (17 September 2022):** cheetah declared extinct 1952 • world's first inter-continental large carnivore translocation • Namibia 8 (Sep 2022) + South Africa 12 (Feb 2023) → **Kuno NP (MP)** • second home Gandhi Sagar Sanctuary (MP, April 2025)
- **International Big Cat Alliance (April 2023):** HQ India • 7 big cats — tiger, lion, leopard, snow leopard, cheetah, jaguar, puma
- **Project Tiger & Elephant (2023):** both projects merged into one division

Timeline of Species Conservation Projects

- **1970** — **Project Hangul (Dachigam, J&K)**
- **1972** — Gir Lion Project (Gujarat)
- **1973** — **Project Tiger — 9 reserves, first Corbett**
- **1975** — Crocodile Conservation Project (Odisha)
- **1977** — **Sangai Project (Keibul Lamjao)**
- **1987** — Project Rhino (Assam)
- **1992** — **Project Elephant (now 33 reserves)**
- **1996** — Red Panda Project (Darjeeling)
- **1999** — **Sea Turtle Project (Olive Ridley)**
- **2006** — Vulture conservation — diclofenac ban
- **2009** — **Project Snow Leopard (5 Himalayan states)**
- **2020** — Project Dolphin (15 Aug)
- **2022** — **Project Cheetah — Kuno (17 Sept)**
- **2023** — International Big Cat Alliance (7 big cats)

Points to Note

1. Elephant — National Heritage Animal (2010) • Tiger — National Animal • Gangetic dolphin — National Aquatic Animal (2009)
2. Project Dolphin in 2020, but dolphin became National Aquatic Animal in 2009
3. Most elephants Karnataka (Assam 2nd, Tamil Nadu 3rd) • most dolphins UP, but highest density Bihar
4. MIKE = CITES programme for monitoring illegal elephant killing • M-STrIPES = tiger monitoring app
5. Rhino: highest number Kaziranga, highest density Pobitora

One-line Facts

- **Indian Rhino Vision 2020 (IRV 2020):** launched 2005 • target — 3,000 one-horned rhinos in 7 protected areas of Assam
- **Salim Ali:** National Park (City Forest) — Srinagar (J&K) • Bird Sanctuary — Chorao Island, Mandovi river (Goa)
- **World-famous pairs:** Dachigam — Hangul • Kaziranga — one-horned rhino • Keoladeo — Siberian crane • Bhitarkanika — saltwater crocodile
- **Silent Valley:** Palakkad (Kerala), Nilgiri hills • **Kunthipuzha river** • main habitat of lion-tailed macaque
- **Valley of Flowers:** Chamoli (Uttarakhand) • Pin Valley — Lahaul-Spiti (Himachal)
- **Mangrove cover:** 1. West Bengal (Sundarbans, about 42%) • 2. Gujarat (Kachchh, about 23%) • 3. Andaman & Nicobar (about 12%)

Part 3: Pollution & Climate Change

07 Environmental Pollution, Diseases & Control Measures

Air Pollutants: Primary & Secondary 14×

- **Primary pollutants (emitted directly from source):** CO • CO2 • SO2 • oxides of nitrogen (NO, NO2) • suspended particulate matter • lead • ammonia • benzene • CFCs
- **Secondary pollutants (formed by reaction of primary pollutants):** **PAN** (peroxyacetyl nitrate) — chief component of photochemical smog • **tropospheric ozone** (VOCs + NOx in sunlight) • sulphuric & nitric acid (acid rain) • SO3 (oxidation of SO2) • smog (smoke + fog)

Feature	London smog (classical)	Los Angeles smog (photochemical)
First episode	London, 1952	Los Angeles, 1943
Main components	Smoke + fog + SO2	O3 + PAN + NOx + aldehydes
Weather	Winter, cool humid morning	Summer, bright sun, warm dry afternoon
Nature	Reducing	Oxidising
Source	Coal/fossil fuel burning	Vehicle & factory emissions

- **Carbon monoxide:** 'asphyxiant gas' • affinity for haemoglobin about **200–250 times** that of O2 → **carboxyhaemoglobin** → blood's O2-carrying capacity lost
- **Sulphur dioxide:** Mathura oil refinery → acid rain → yellowing of Taj Mahal (**stone leprosy / marble cancer**) • protected by Taj Trapezium Zone (TTZ, about 10,400 sq km) • best bio-indicator **lichens** (fungus + alga)
- **Fly ash:** fine powder from coal-based thermal power plants • constituents: **silica (SiO2)**, aluminium silicate, CaO + heavy metals (lead, arsenic, cobalt, copper) • uses: bricks • Portland cement • road embankments • raising soil water-holding capacity
- **Lead:** anti-knocking agent **tetraethyl lead** in petrol • damages central nervous system & brain • unleaded petrol nationwide in India since 2000
- **Benzene:** cigarette smoke, paint, petrol • **blood cancer (leukaemia)**

Points to Note

1. Ozone: life-saving shield in the stratosphere; in the troposphere a **secondary pollutant** & greenhouse gas
2. PAN & tropospheric ozone are not emitted directly — both secondary
3. London smog = winter, SO2, **reducing**; Los Angeles = summer, O3/PAN, **oxidising** — examiners swap them

Air Quality Index (AQI) & Control Programmes

- **National AQI:** launched **2014** • principle 'One Number, One Colour, One Description' • **6 categories**
- **AQI categories:** Good 0–50 (dark green) • Satisfactory 51–100 (light green) • Moderately polluted 101–200 (yellow) • Poor 201–300 (orange) • Very poor 301–400 (red) • **Severe 401–500** (maroon)
- **8 pollutants:** PM10 • PM2.5 • NO2 • SO2 • CO • O3 • NH3 • lead • highest sub-index = AQI • minimum 3 pollutants, one PM mandatory
- **NAAQS 2009: 12 pollutants** — the 8 of AQI + benzene, benzo(a)pyrene, arsenic, nickel • PM2.5 40 (annual)/60 (24-hr) • PM10 60/100 µg/m^3
- **WHO guidelines 2021:** PM2.5 5 (annual)/15 (24-hr) • PM10 15/45 µg/m^3
- **SAFAR: Ministry of Earth Sciences** • developed by IITM Pune • launched 2010 (Commonwealth Games, Delhi) • also monitors PM1, black carbon, **benzene, toluene, xylene, mercury**

- **National Clean Air Programme (NCAP): 10 January 2019** • Environment Ministry • initial target 20–30% cut by 2024 (base 2017) • revised **up to 40% cut in PM10 by 2025-26** • about 130 cities • portal PRANA
- **GRAP:** Delhi-NCR • since 2017 • 4 stages — Poor • Very poor • Severe • Severe+ (above 450) • enforced by **Commission for Air Quality Management (CAQM, 2021)**

Air Quality Index (AQI)

0–50 Good	51–100 Satisfactory	101–200 Moderate	201–300 Poor	301–400 Very Poor	401–500 Severe

8 pollutants used in AQI:

PM10 · PM2.5 · NO_2 · SO_2 · CO · O_3 · NH_3 · Pb

National AQI: 17 Oct 2014 (Swachh Bharat) · CPCB · 24-hour average

Points to Note

1. AQI includes **CO, not CO2** • benzene is not in AQI, it is in NAAQS
2. AQI = 8 pollutants, 6 categories; NAAQS = 12 pollutants
3. SAFAR = **Ministry of Earth Sciences**; AQI & NCAP = Environment Ministry/CPCB

Water Pollution & Water-Quality Indicators 11×

- **Dissolved oxygen (DO):** needed for respiration of aquatic life • clean water about **8 mg/L** or more • **below 4 mg/L** → highly polluted, fish die • falls as temperature rises
- **Biochemical oxygen demand (BOD):** O2 needed by aerobic microbes to decompose only **biodegradable** organic matter in **5 days at 20°C**
- **Chemical oxygen demand (COD):** O2 needed to chemically oxidise **both** biodegradable + non-biodegradable matter • oxidant potassium dichromate
- **Golden rule:** always **COD > BOD** • as pollution rises BOD & COD **rise**, DO **falls**
- **CPCB bathing-water criteria:** DO ≥ 5 mg/L • BOD ≤ 3 mg/L • faecal coliform ≤ 2,500 MPN/100 ml
- **Eutrophication:** excess nitrate & phosphate (fertiliser, sewage) → **algal bloom** → DO exhausted → **'biological desert'** • water hyacinth = 'Terror of Bengal'
- **Biomagnification:** progressive rise in concentration of non-degradable pollutants (DDT, mercury, endosulfan, strontium-90) at higher trophic levels • maximum in **top carnivores** (vulture, hawk, humans) • DDT: water 0.003 ppb → fish-eating birds 25 ppm → thin eggshells
- **Diclofenac:** veterinary painkiller • vultures eating carcasses of treated cattle suffer **kidney failure** → steep decline • veterinary use banned 2006
- **Polluted rivers:** Damodar — 'biological desert' • UP: **Hindon** (Saharanpur → Ghaziabad-Noida) & **Gomti** (Lucknow–Jaunpur) — very high BOD, 'biological desert'-like state
- **Kali river (East):** Muzaffarnagar–Meerut to the Ganga near Kannauj • sugar mill & paper industry waste • Ramganga & Sai also polluted by industrial waste

Points to Note

1. BOD counts only **biodegradable** matter; COD counts **both** — hence COD is always higher
2. As pollution rises BOD-COD rise, but **DO falls**
3. Standard BOD condition: **20°C, 5 days**
4. Bioaccumulation = within one organism over time; biomagnification = across trophic levels

Wastewater (Sewage) Treatment

- **Primary (physical) treatment:** screening • grit removal • sedimentation → **primary sludge**
- **Secondary (biological) treatment:** in the **aeration tank** aerobic bacteria form **flocs** (bacteria + fungal filaments) that consume organic matter → **BOD sharply reduced** • settling gives **activated sludge**; part returned as inoculum
- **Anaerobic sludge digester:** remaining sludge digested → **biogas** (methane + H2S + CO2)
- **Tertiary treatment:** nitrogen-phosphorus removal • disinfection (chlorine, ozone, UV)
- **Natural options:** oxidation ponds • constructed wetlands • **East Kolkata Wetlands** (Ramsar 2002) — natural sewage treatment

Stages of Sewage Treatment

Primary	Secondary	Tertiary
Physical	Biological	Chemical/advanced
Screening Grit removal Sedimentation → primary sludge	Aerobic microbes Activated sludge (flocs) cut BOD sharply	Remove N, P (stops eutrophication) Disinfection by chlorine/UV

Anaerobic sludge digester: secondary sludge → biogas (mainly CH_4 + CO_2, H_2S)
High BOD = more organic pollution · clean water BOD < 5 mg/L

Points to Note

1. Primary stage is purely **physical**; BOD falls most in the **secondary** stage
2. Aeration tank = **aerobic**; biogas = **anaerobic** digester

Major Pollution-Related Diseases & Syndromes 16×

- **Mercury → Minamata disease:** nervous system damage • Minamata Bay, Japan (1956), from contaminated fish • Minamata Convention 2013 (in force 2017)
- **Cadmium → Itai-itai ('ouch-ouch'):** severe bone & joint pain, kidney damage • Toyama Prefecture, Japan (about 1912)
- **Nitrate → Blue baby syndrome (methaemoglobinaemia):** nitrate above **45 mg/L** in drinking water • O2 transport blocked in infants, body turns blue
- **Arsenic → Black foot disease, keratosis:** cracked skin, black spots on soles, skin & lung cancer • Ganga-Brahmaputra plains — Ballia, Ghazipur, West Bengal • drinking-water limit 0.01 mg/L
- **Fluoride → Fluorosis (knock-knee syndrome):** yellow/mottled teeth, bent bones, knees knocking • Unnao, Agra, Mathura • drinking-water limit 1.0 (max 1.5) mg/L
- **Asbestos → Asbestosis:** fibrosis of lung lining, **mesothelioma** cancer
- **Coal dust → Anthracosis ('black lung'):** coal miners
- **Silica dust → Silicosis:** stone-crushing & glass industry workers
- **Iron particles → Siderosis:** iron-ore mining
- **Cotton dust → Byssinosis ('brown lung'):** textile mill workers
- **Sugarcane bagasse dust → Bagassosis:** sugar mill workers

Points to Note

1. Minamata = **mercury**; Itai-itai = **cadmium** — both Japanese, often swapped in options
2. Blue baby = nitrate; CO poisoning = carboxyhaemoglobin — both cut O2 transport

3. Black foot = arsenic; black lung = coal; brown lung = cotton
4. Silicosis = silica; Siderosis = iron

Noise & Radioactive Pollution, Industrial Disasters 8×

- **Noise measurement:** intensity — **decibel** • frequency — **hertz**
- **WHO:** safe level **45 dB** • above **80 dB** = noise (pollution)
- **Noise Pollution (Regulation and Control) Rules 2000:** day = 6 AM to 10 PM • silence zone = **100 m** around hospitals, educational institutions, courts

Zone	Day (dB)	Night (dB)
Industrial	75	70
Commercial	65	55
Residential	55	45
Silence zone	50	40

- **Green muffler:** **noise pollution** control • 4–6 rows of dense trees like **Ashoka, Neem** along roads, highways & industrial areas → absorb sound waves
- **Radioactive pollutants & target organs:** strontium-90 → bones • iodine-131 → thyroid • caesium-137 → soft tissue • radon → lung cancer • units: becquerel, curie (activity) • sievert, rem (biological effect)
- **Bhopal gas tragedy:** **2–3 December 1984** • Union Carbide India Ltd. pesticide plant • **methyl isocyanate (MIC)** leak • led to Environment (Protection) Act 1986
- **Chernobyl nuclear accident:** **26 April 1986** • Ukraine (then USSR)
- **Fukushima Daiichi nuclear accident:** **11 March 2011** • Japan • caused by tsunami

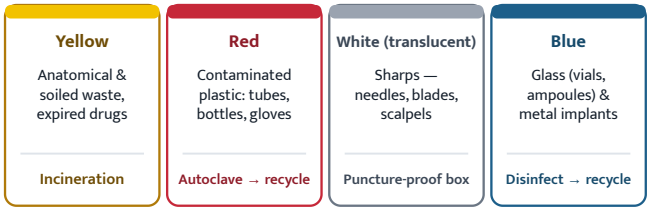
Points to Note

1. Bhopal = methyl **isocyanate** — 'isocyanide' option is wrong
2. **Green muffler** = noise pollution control; nothing to do with afforestation or greenhouse
3. Lowest limit is silence zone (50/40); industrial day-night gap only 5 dB
4. Decibel is a logarithmic scale: +10 dB = 10 times intensity

Solid Waste, E-waste, Plastic & Bioremediation

- **Solid Waste Management Rules 2026:** in force **1 April 2026** (replacing 2016 rules) • urban & rural both • **4-stream segregation** at source — wet • dry • sanitary • special care (bulbs, batteries, medicines) • bulk waste generators (≥ 20,000 sq m area or ≥ 100 kg/day waste) carry special responsibility • landfill only for non-recyclable/inert waste
- **E-Waste (Management) Rules 2022:** in force **1 April 2023** • earlier rules 2011, 2016 • 106 electrical-electronic items • solar PV panels included • **EPR** (Extended Producer Responsibility) on producers, online EPR certificates
- **Plastic Waste Management Rules 2016:** ban on identified **single-use plastic** from **1 July 2022** • carry bag 75 micron (September 2021) → **120 micron** (December 2022) • 2024 amendment — definition of biodegradable plastic
- **Other rules:** Bio-medical Waste 2016 • Hazardous Waste 2016 • Battery Waste 2022 • Construction & Demolition (C&D) Waste 2025 (in force 1 April 2026)
- **Bio-medical waste colour code:** yellow — anatomical, infectious waste (incineration) • red — recyclable plastics (autoclave) • white — needles, sharps • blue — glassware & metallic implants
- **Oilzapper:** **TERI** • mix of **5 bacteria** • converts hydrocarbons of marine oil spills into CO2 & water • companion product Oilivorous-S
- **Superbug (Pseudomonas putida):** Dr. **Ananda Mohan Chakrabarty** • oil-eating bacterium made by genetic engineering • first GM organism to be patented (USA, 1980)
- **Phytoremediation:** plants absorb heavy metals from soil/water • sunflower, water hyacinth, mustard

Bio-Medical Waste Colour Code



Bio-Medical Waste Management Rules 2016 — 4 colour categories

Points to Note

1. SWM Rules 2026 = **4 streams**; 2016 = 3 streams (wet, dry, domestic hazardous)
2. E-waste Rules 2022, but in force from **2023**; C&D Rules 2025, in force from **2026**
3. Current minimum carry-bag thickness **120 micron**
4. Superbug = **Pseudomonas putida** (Chakrabarty); Oilzapper = **TERI**

One-line Facts

- **Main pollutants in cigarette smoke:** CO • benzene
- **Disinfection of drinking water:** chlorine gas • bleaching powder • ozone
- **Polythene & plastic:** not decomposed by ordinary bacteria → **non-biodegradable**
- **Acid rain:** pH **below 5.6** • sulphuric acid about 70% • nitric acid about 30%
- **International Acid Rain Information Centre:** Manchester (UK) • Scandinavian countries suffer most damage
- **National Pollution Control Day:** 2 December (Bhopal tragedy remembrance)

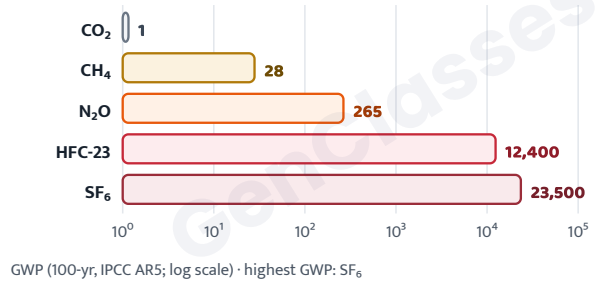
08 Climate Change, Global Warming & Ozone Depletion

Greenhouse Gases & Global Warming Potential 18×

- **Concept:** **Joseph Fourier (1824)** • **Svante Arrhenius (1896)** — CO2 added by burning fossil fuels warms the Earth
- **Mechanism:** transparent to incoming **short-wave solar radiation** • absorb the **long-wave infrared (terrestrial) radiation** going back from Earth → atmosphere warms
- **Natural greenhouse effect:** without it Earth's mean temperature ≈ −18°C, with it ≈ 15°C • the problem = human-made "enhanced" effect
- **Global Warming Potential (GWP):** heat trapped by 1 kg of a gas over 100 years, relative to 1 kg of CO2 • **CO2 = 1** (reference)
- **Water vapour:** **highest (≈60%)** in amount and warming contribution • not counted in GWP • lifetime 9–10 days • oceanic evaporation

- **Carbon dioxide: largest share of human-made warming (≈60%)** • GWP 1 • lifetime 100–300 years • fossil fuels, deforestation, cement • pre-industrial ≈280 ppm → now above 420 ppm
- **Methane (marsh gas):** ≈20% • GWP **28** • lifetime **≈12 years** • paddy fields, ruminant livestock, wetlands, coal mines, termites, landfills
- **Nitrous oxide (laughing gas):** ≈6% • GWP **265** • lifetime 114–121 years • nitrogen fertilisers, nylon production, biomass burning
- **Chlorofluorocarbons (CFC):** ≈14% (with other halocarbons) • GWP ≈4,660 (CFC-11) to ≈10,200 (CFC-12) • lifetime 45–100 years • refrigerators, AC, foam
- **Sulphur hexafluoride (SF6):** very small quantity • GWP **23,500 (highest)** • lifetime 3,200 years • electrical transformers, magnesium production, semiconductors
- **Others:** HFC-23 GWP 12,400 • perfluorocarbon CF4 lifetime ≈50,000 years (longest-lived)
- **6 gases controlled under Kyoto:** CO2 • methane • nitrous oxide • hydrofluorocarbons (HFC) • perfluorocarbons (PFC) • sulphur hexafluoride (SF6) • 7th **nitrogen trifluoride** — Doha Amendment 2012

Global Warming Potential of GHGs



GWP (100-yr, IPCC AR5; log scale) · highest GWP: SF_6

Points to Note

1. Most abundant greenhouse gas naturally = **water vapour** • largest and most important from human emissions = **CO2**
2. Highest GWP = SF6 • longest-lived = CF4 — two different gases
3. GWP in older books: methane 21/25, nitrous oxide 310/298 • current standard (IPCC AR5) 28 and 265
4. Water vapour and CFCs are not in the Kyoto list — CFCs fall under the Montreal Protocol

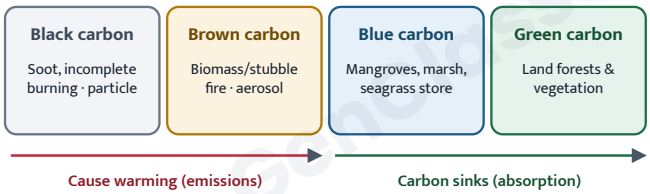
Black Carbon, Brown Carbon & Blue Carbon 6×

- **Black carbon (soot):** **particulate pollutant** (not a gas) formed by **incomplete combustion** of fossil fuels, biofuels and biomass
- **Effect of black carbon:** strong absorber of sunlight • settles on Himalayan/Arctic glaciers and **lowers the albedo (reflectivity)** of snow → faster melting • lifetime only **a few days to a few weeks**
- **Brown carbon:** **organic aerosol** from burning biomass (wood, crop residue/stubble) • absorbs mainly ultraviolet and blue light
- **Blue carbon:** carbon stored by ocean and coastal ecosystems — **mangroves, tidal salt marshes, seagrasses** • absorbs **≈10 times** faster per unit area than terrestrial forests • most of the stock lies in sediment
- **India's mangrove initiatives:** MISHTI (Budget 2023-24) • India is a member of the Mangrove Alliance for Climate (MAC, COP27, 2022)

Points to Note

1. Black carbon is not a gas but particulate matter — not among the Kyoto gases, yet a strong warming agent
2. Black carbon = incomplete combustion (fossil + bio) • brown carbon = mainly biomass/stubble burning
3. Blue carbon ecosystems = mangroves, salt marshes, seagrasses • coral reefs not included

The "Colours" of Carbon



Ozone Layer & Ozone Depletion 15×

- **Location: 90% of total ozone in the stratosphere** (15–35 km; highest concentration at **20–25 km**) • remaining 10% in the troposphere (a pollutant and greenhouse gas there)
- **Measurement:** Dobson Unit (DU) • 100 DU = **1 mm** thick layer of pure ozone at 0°C and 1 atmosphere pressure • average ≈300 DU • **below 220 DU = ozone hole**
- **Discovery:** ozone layer — **Charles Fabry & Henri Buisson** (1913, France) • spectrophotometer and ozone studies — **G.M.B. Dobson**
- **Discovery of the Antarctic ozone hole: 1985** • Joseph Farman, Brian Gardiner, Jonathan Shanklin (British Antarctic Survey)
- **Ozone-depleting substances:** CFCs (freon) • halons (fire extinguishers) • carbon tetrachloride • methyl chloroform • methyl bromide • **nitrous oxide** (supersonic jets)
- **Ozone Depletion Potential (ODP):** CFC-11 = 1 (reference) • halons highest • HCFCs low (interim substitutes) • HFC = **0**
- **Active chlorine:** UV action on CFCs → free **chlorine atom** • 1 chlorine atom destroys ≈ **1 lakh (100,000)** ozone molecules • bromine even more destructive per atom
- **Chlorine catalytic cycle:** Cl + O3 → ClO + O2 • ClO + O → Cl + O2 • chlorine is freed again and acts repeatedly
- **Polar Stratospheric Clouds (PSC):** form in the Antarctic **winter (June–August)** below −78°C inside the **polar vortex** • on their surface inactive chlorine → active chlorine
- **Timing of the ozone hole: spring (September–November)** — rapid depletion as sunlight returns • maximum September–October • fills up when the polar vortex breaks
- **Why the biggest hole over Antarctica:** colder, strong polar vortex, more PSCs • Arctic relatively warmer → hole smaller and occasional
- **Recovery (UNEP–WMO Assessment 2023):** to 1980 levels — rest of the world ≈2040 • Arctic ≈2045 • Antarctica ≈2066

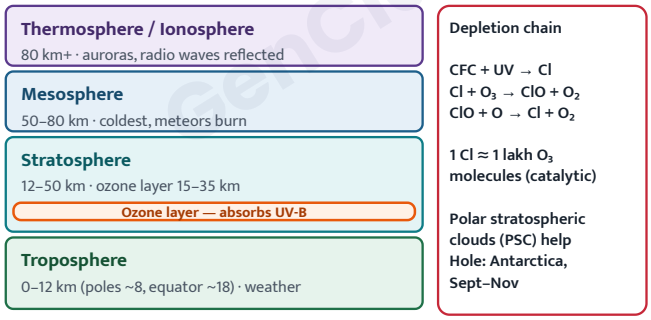
Ultraviolet ray	Absorption by ozone	Effect
UV-A (315–400 nm)	No — reaches the surface	Least harmful; wrinkles, skin ageing
UV-B (280–315 nm)	Partial	Skin cancer (melanoma), cataract, DNA mutation
UV-C (100–280 nm)	Complete	Most lethal; germicidal

Points to Note

1. Ozone hole is over **Antarctica** in **spring (September–November)** — PSCs form in winter, depletion starts when sunlight returns

2. Ozone hole = severe thinning of ozone (below 220 DU), not an actual hole in the layer
3. Nitrous oxide is today the largest ozone-depleting emission — yet not controlled under Montreal
4. 100 DU = 1 mm (not 1 DU = 1 mm) • the most lethal UV-C does not reach the surface

Atmospheric Layers & Ozone Depletion



Unit: Dobson Unit (DU) · below 220 DU = ozone hole · Ozone Day 16 Sept

Ozone Protection: Vienna, Montreal & Kigali

- **Vienna Convention: 22 March 1985** • framework agreement for ozone protection (no binding targets) • in force 1988 • India joined 1991
- **Montreal Protocol:** signed **16 September 1987** • in force **1 January 1989** • binding phased elimination of ozone-depleting substances • India joined 1992 • with Vienna, the first universally ratified treaty in UN history (2009)
- **Kigali Amendment: 15 October 2016**, Kigali (Rwanda) • phase-down of HFCs • in force 1 January 2019 • India ratified in 2021
- **India under Kigali:** baseline 2024–26 • freeze 2028 • reduction begins **2032** • **85% by 2047**
- **India:** phase-out of CFCs, halons, carbon tetrachloride completed by 2010 • **India Cooling Action Plan (ICAP, March 2019)**
- **World Ozone Day: 16 September** (date of signing of the Montreal Protocol) • declared by UN General Assembly 1994

Points to Note

1. Vienna (1985) = framework, no binding targets • Montreal (1987) = binding elimination • Montreal in force 1989
2. HFCs have zero ODP but are potent greenhouse gases → Kigali Amendment is for HFCs, not CFCs
3. Kigali (Rwanda) ≠ Kyoto (Japan) • Kigali amends the Montreal Protocol, not the UNFCCC
4. India's HFC reduction from 2032 — developed countries from 2019

National Action Plan on Climate Change (NAPCC) 9×

- **Launch: 30 June 2008** • released by the Prime Minister's Council on Climate Change • **8 National Missions**
- **National Solar Mission:** Jawaharlal Nehru National Solar Mission (2010) • Ministry of New & Renewable Energy • target 20 GW → **100 GW** in 2015 (by 2022)
- **National Mission for Enhanced Energy Efficiency:** Ministry of Power/Bureau of Energy Efficiency • **PAT** (Perform, Achieve and Trade) scheme
- **National Mission on Sustainable Habitat:** Ministry of Housing & Urban Affairs • buildings, urban waste, public transport
- **National Water Mission:** Ministry of Jal Shakti • **20%** increase in water-use efficiency
- **National Mission for Sustaining the Himalayan Ecosystem:** Department of Science & Technology (DST)
- **National Mission for a Green India (Green India Mission):** Environment Ministry • 10 million ha (5 forest-cover increase + 5 forest-quality improvement)
- **National Mission for Sustainable Agriculture:** Ministry of Agriculture & Farmers Welfare • climate-resilient farming
- **National Mission on Strategic Knowledge for Climate Change:** Department of Science & Technology (DST)

Points to Note

1. **Wind Energy Mission and Coastal Mission** are not among the original 8 missions
2. Himalayan Ecosystem and Strategic Knowledge — both missions under DST, not the Environment Ministry
3. NAPCC in 2008, Solar Mission launched in 2010 • PAT falls under the Energy Efficiency Mission

India's Climate Targets: Panchamrit & NDC

- **Panchamrit:** 5 targets announced by PM Modi at **COP26 Glasgow (November 2021)**
- **Target 1 — 500 GW:** non-fossil energy capacity of 500 gigawatts by 2030
- **Target 2 — 50%:** 50% of energy requirement from renewables by 2030 • in NDC: 50% of **installed electricity capacity** non-fossil — **already achieved in June 2025**
- **Target 3 — 1 billion tonnes:** cut total projected carbon emissions by 1 billion tonnes from now to 2030
- **Target 4 — 45%:** cut emission intensity by 45% from 2005 level by 2030
- **Target 5 — Net zero: by 2070**

Target	2015 → 2022 update (by 2030)	NDC 3.0 (by 2035)
Emission intensity cut (base 2005)	33–35% → 45%	47%
Non-fossil installed power capacity	40% → 50%	60%
Carbon sink (forest & tree cover)	2.5–3 billion t CO2e	3.5–4 billion t CO2e

- **NDC 3.0:** Union Cabinet approval **25 March 2026** • period 2031–35
- **Mission LiFE (Lifestyle for Environment):** proposed at COP26 (2021) • launched **20 October 2022, Kevadia (Gujarat)** • "mindful utilisation" in place of "destructive consumption" • Pro-Planet People
- **Green Credit Programme:** notified 12 October 2023 (Environment Protection Act 1986) • voluntary market-based mechanism • administrator ICFRE

Points to Note

1. NDC's 50% = share of non-fossil **installed capacity** — not of actual power generation
2. Intensity target = cut in emissions per unit of GDP, not in total emissions • base year 2005
3. 2030 targets 45% and 50% • 2035 (NDC 3.0) targets 47% and 60%
4. 500 GW and 1 billion tonnes — announced in Panchamrit, but not in the formal 2022 NDC
5. Net zero: India **2070** • China 2060 • European Union 2050

India's Climate Targets

Panchamrit — COP26 Glasgow (2021)

1. 500 GW non-fossil capacity (2030)
2. 50% energy from renewables (2030)
3. 1 billion t emission cut (2030)
4. Emission intensity −45% (2030)
5. Net Zero by 2070

India's NDC journey

	2015 NDC target 2030	2022 update target 2030	NDC 3.0 (2026) target 2035
Emission intensity↓	33–35%	45%	47%
Non-fossil capacity	40%	50% ✓ June 2025	60%
Carbon sink	2.5–3 bn t	2.5–3 bn t	3.5–4 bn t

Base year 2005 · 50% non-fossil installed capacity reached 5 years early

One-line Facts

- **Ozone halo:** over the Tibetan Plateau • discovered 2005, G.W. Kent Moore
- **Carbon-negative countries:** **Bhutan** • **Suriname** • Panama — absorb more than they emit
- **Earth Hour:** WWF • started 2007, Sydney (Australia) • every year on the **last Saturday of March**, 8:30–9:30 pm
- **Carbon credit:** concept from the **Kyoto Protocol (1997)** • **1 carbon credit** = **1 tonne of CO2** (or equivalent) emission reduced
- **Biochar:** charcoal made by **pyrolysis** of farm waste (thermal decomposition at high temperature without oxygen) • carbon sequestration in soil

Part 4: International Conventions, Indian Laws & Institutions

09 International Environmental Conventions, Treaties & Organisations

Major Global Environmental Conventions: Chronology 24×

- **Ramsar Convention (1971):** Ramsar (Iran) • signed 2 February 1971, in force **1975** • conservation of wetlands • India joined 1 February 1982
- **Stockholm Conference (1972):** UN Conference on the Human Environment • began 5 June 1972 (World Environment Day) • **'Magna Carta'** of environment • outcome: **UNEP** • India represented by Indira Gandhi (only head of government apart from the host)
- **World Heritage Convention (1972):** Paris, UNESCO • protection of natural & cultural heritage • in force 1975 • India 1977
- **CITES (1973):** Washington DC • international trade in endangered species of wild fauna & flora • in force **1975** • 3 Appendices • **legally binding** • India 1976
- **Bonn Convention / CMS (1979):** Bonn (Germany) • conservation of migratory species of wild animals • in force 1983
- **Vienna Convention (1985):** first legal framework for protecting the **ozone layer** • in force 1988 • no binding targets
- **Brundtland Commission Report (1987):** World Commission on Environment and Development (chair Gro Harlem Brundtland, Norway) • title **'Our Common Future'** • most accepted definition of sustainable development
- **Montreal Protocol (1987):** Montreal (Canada) • signed **16 September 1987** (World Ozone Day), in force 1 January 1989 • controls ozone-depleting substances (CFCs, halons) • first UN treaty with universal ratification • India 1992
- **Basel Convention (1989):** Basel (Switzerland) • **transboundary movement** and disposal of hazardous wastes • in force 1992
- **Earth Summit (1992):** Rio de Janeiro • 20th anniversary of Stockholm • UNFCCC (in force 1994) and CBD (in force 1993) opened for signature
- **UNCCD (1994):** Paris • combating desertification • in force 1996
- **Kyoto Protocol / COP-3 (1997):** Kyoto (Japan) • adopted 11 December 1997, in force **16 February 2005** • **legally binding** GHG cuts, only for developed (Annex-I) countries • first period 2008–12 • 6 gases • India 2002 (no binding target)
- **Rotterdam Convention (1998):** **Prior Informed Consent (PIC)** procedure for trade in hazardous chemicals & pesticides • in force 2004
- **Cartagena Protocol (2000):** adopted in Montreal • supplement to CBD • **biosafety** — safe transfer of living modified organisms (LMO/GMO) • in force 2003
- **Stockholm Convention (2001):** **Persistent Organic Pollutants (POPs)** — initial 12 'Dirty Dozen' • in force 2004
- **Rio+10 (2002):** World Summit on Sustainable Development, Johannesburg • review of Agenda 21 • reaffirmed Millennium Development Goals
- **Nagoya Protocol / CBD COP-10 (2010):** Nagoya (Japan) • **Access and Benefit Sharing (ABS)** of genetic resources • in force 2014 • same COP set 20 Aichi Targets (2011–20)
- **Rio+20 (2012):** Rio de Janeiro • outcome document **'The Future We Want'** • launched SDGs and green economy idea
- **Minamata Convention (2013):** signed at Kumamoto (Japan) • protection from **mercury** • in force 2017 • named after Minamata disease
- **Paris Agreement / COP-21 (2015):** adopted 12 December 2015, in force **4 November 2016** • limit warming well below 2°C over pre-industrial, ideally **1.5°C** • India ratified 2 October 2016 • USA out for the second time (effective January 2026)
- **Kigali Amendment (2016):** Kigali (Rwanda) • to the Montreal Protocol • **HFCs** (ozone-safe but potent GHGs) cut 80–85% by 2047 • in force 1 January 2019 • binding • India 2021: phase-down from 2032, 85% by 2047
- **Kunming-Montreal Global Biodiversity Framework / CBD COP-15 (2022):** Montreal • 4 goals, 23 targets • **'30x30'** — protect 30% of land and 30% of oceans by 2030
- **BBNJ / High Seas Treaty (2023):** marine biodiversity beyond national jurisdiction • in force **17 January 2026**

Chronology of Major Environmental Conventions

1971	Ramsar (wetlands)	1997	Kyoto Protocol
1972	Stockholm Conf. → UNEP	1998	Rotterdam (PIC)
1973	CITES	2000	Cartagena (biosafety)
1979	CMS (Bonn)	2001	Stockholm (POPs)
1985	Vienna (ozone)	2010	Nagoya (ABS)
1987	Montreal Protocol	2013	Minamata (mercury)
1988	IPCC	2015	Paris Agreement
1989	Basel (hazardous waste)	2016	Kigali Amendment (HFC)
1992	Rio: UNFCCC, CBD	2022	Kunming-Montreal GBF
1994	UNCCD	2025	COP30 Belém (Brazil)

Year = adopted/signed; entry-into-force year may differ

Points to Note

1. Ramsar (signed 1971) and CITES (signed 1973) — both came into force in 1975
2. Vienna = only a framework • Montreal = binding controls
3. Stockholm Conference (1972, human environment) ≠ Stockholm Convention (2001, POPs)
4. Cartagena Protocol adopted in Montreal • Minamata Convention signed at Kumamoto
5. Kyoto: adopted 1997, in force 2005 • Paris: adopted 2015, in force 2016

Climate Change Conferences (UNFCCC COP) & Latest Summits 12×

- **COP-1 (1995):** **Berlin** (Germany)
- **COP-3 (1997):** Kyoto Protocol • 3 flexible mechanisms: Clean Development Mechanism (CDM) • Joint Implementation (JI) • Emissions Trading (carbon trading)
- **COP-8 (2002):** **New Delhi** • the only UNFCCC COP held in India
- **COP-13 (2007):** Bali (Indonesia) • Bali Action Plan
- **COP-15 (2009):** Copenhagen (Denmark) • developed countries pledged **USD 100 billion** per year by 2020
- **COP-16 (2010):** Cancún (Mexico) • formal establishment of the **Green Climate Fund (GCF)** • HQ Songdo, Incheon (South Korea)
- **Other COPs:** COP-18 Doha (2012) — Kyoto second period 2013–20 • COP-19 Warsaw (2013) — Warsaw International Mechanism on Loss & Damage (WIM)
- **COP-21 (2015):** Paris • **Nationally Determined Contributions (NDCs)** • International Solar Alliance (ISA) launched by India & France, HQ **Gurugram**
- **COP-26 (2021):** Glasgow (UK) • India's Panchamrit (net zero by 2070) • coal 'phase-out' dropped for **'phase-down'**
- **COP-27 (2022):** Sharm el-Sheikh (Egypt) • agreement on **'Loss and Damage Fund'** • India's Long-Term Low Emission Development Strategy
- **COP-28 (2023):** Dubai (UAE) • 'UAE Consensus' • first **Global Stocktake** • Loss & Damage Fund operationalised (World Bank interim trustee) • tripling renewable capacity and doubling energy efficiency by 2030 • 'transitioning away' from fossil fuels • India: Green Credit Initiative
- **COP-29 (2024):** **Baku (Azerbaijan)** • 'Finance COP' • New Collective Quantified Goal (NCQG): **USD 300 billion** per year by 2035, overall goal 1.3 trillion • Paris Article 6 (carbon market rules) finalised
- **COP-30 (10–22 November 2025):** **Belém (Brazil)** • first COP in the Amazon • 'Global Mutirão' decision • call to triple adaptation finance by 2035 • Belém indicators for the Global Goal on Adaptation • **Tropical Forests Forever Facility** launched • Just Transition mechanism • no fossil-fuel roadmap in the text
- **COP-31 (9–20 November 2026):** **Antalya (Türkiye)** • Türkiye hosts and presides • Australia leads negotiations • pre-COP in the Pacific • COP-32 (2027) Addis Ababa (Ethiopia)
- **CBD COPs:** COP-16 **Cali (Colombia)**, October 2024 — 'Cali Fund' for benefit sharing from Digital Sequence Information, permanent subsidiary body for Indigenous Peoples • COP-17 **Yerevan (Armenia)**, 19–30 October 2026 — first global review of the KMGBF
- **Other latest COPs:** Ramsar COP-15 Victoria Falls, Zimbabwe (July 2025) • CITES CoP20 Samarkand, Uzbekistan (November–December 2025; 50 years of CITES) • CMS COP-14 Samarkand (February 2024) • UNCCD COP-16 Riyadh (December 2024), COP-17 Ulaanbaatar, Mongolia (August 2026)

Recent Major Environmental Summits



1 COP26 Glasgow 2021 · 2 COP27 Sharm el-Sheikh 2022 · 3 COP28 Dubai 2023 · 4 COP29 Baku 2024 · 5 COP30 Belém 2025 · 6 COP31 Antalya 9–20 Nov 2026 · 7 CBD COP16 Cali 2024 · 8 CBD COP17 Yerevan 2026 · 9 Ramsar COP15 Victoria Falls 2025 · 10 CITES CoP20 Samarkand 2025 (orange/red = UNFCCC · green = CBD · blue = Ramsar · purple = CITES)

Points to Note

1. Only one UNFCCC COP in India — COP-8 New Delhi (2002) • others: CBD COP-11 Hyderabad (2012) • UNCCD COP-14 New Delhi (2019) • CMS COP-13 Gandhinagar (2020)
2. Loss & Damage Fund: agreed at Sharm el-Sheikh (2022) • operationalised at Dubai (2023)
3. GCF: USD 100 billion pledge at Copenhagen (2009) • formally set up at Cancún (2010)
4. Glasgow = coal 'phase-down' • no agreement on 'phase-out'
5. COP-31 in Türkiye, but negotiations led by Australia • CITES CoP20 and CMS COP-14 both in Samarkand

Agenda 21 & Sustainable Development Goals (SDGs) 14×

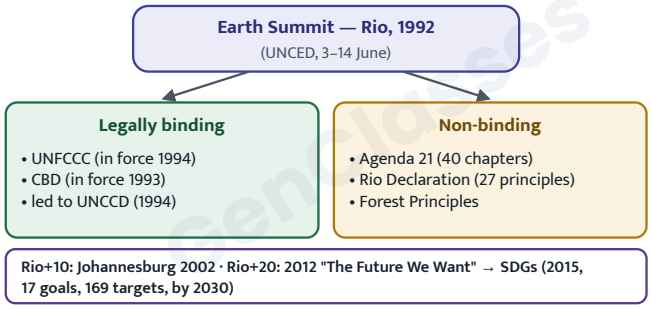
- **Rio Earth Summit (1992):** 5 outcomes: Agenda 21 • Rio Declaration (27 principles) • Forest Principles (non-binding) • UNFCCC • CBD • three Rio sisters = UNFCCC, CBD, UNCCD
- **Key Rio Declaration principles:** 7 = Common But Differentiated Responsibilities (CBDR) • 15 = precautionary approach • 16 = polluter pays
- **Agenda 21:** non-binding, voluntary action plan for sustainable development in the **21st century** • **4 sections, 40 chapters**
- **Sustainable Development Goals (SDGs):** UN General Assembly, 2015 • **'Agenda 2030'** • effective 1 January 2016 to 2030 • **17 goals, 169 targets** • predecessor: Millennium Development Goals (8 goals, 2000–15)
- **Environment-linked SDGs:** 6 Clean water & sanitation • 7 Affordable & clean energy • 11 Sustainable cities & communities • 12 Responsible consumption & production
- **SDG 13:** **Climate Action**
- **SDG 14:** **Life Below Water** — ocean conservation
- **SDG 15:** **Life on Land** — forests, desertification, biodiversity

Points to Note

1. Agenda 21 has **40 chapters** — '21' = 21st century • it is non-binding
2. SDG 14 = marine life • SDG 6 = clean water • forests & desertification = SDG 15
3. Term 'sustainable development' first used in the World Conservation Strategy (1980, IUCN) • most accepted definition from the Brundtland Report (1987)

4. Rio+5 New York (1997) • Rio+10 Johannesburg (2002) • Rio+20 Rio de Janeiro (2012)

Outcomes of the Rio Earth Summit



Major International Environmental Organisations & Headquarters 10×

- **IUCN (International Union for Conservation of Nature): 1948 • Gland (Switzerland)** • world's oldest global environmental network • Red List since 1964
- **WWF (World Wide Fund for Nature): 1961 • Gland (Switzerland)** • logo Giant Panda • Living Planet Report • Earth Hour
- **UNEP: 1972 • Nairobi (Kenya)** • reports: Emissions Gap, Adaptation Gap, Global Environment Outlook • awards: Champions of the Earth, Global 500 (1987–2003)
- **IPCC (Intergovernmental Panel on Climate Change): 1988 • Geneva (Switzerland)** • set up by UNEP + WMO • 2007 Nobel Peace Prize (shared with Al Gore; then chair R.K. Pachauri) • current chair Jim Skea (since 2023)
- **Greenpeace International: 1971 • Amsterdam (Netherlands)** • NGO founded in Vancouver (Canada)
- **Wetlands International:** origin 1937, present name 1995 • **Ede (Netherlands)** (earlier Wageningen) • conservation & restoration of wetlands
- **BirdLife International: 1922 • Cambridge (UK)** • bird conservation • declares Important Bird and Biodiversity Areas (IBAs) • India partner BNHS
- **TRAFFIC: 1976 • Cambridge (UK)** • wildlife trade monitoring network • joint programme of WWF and IUCN
- **Global Environment Facility (GEF): 1991 • Washington DC** • set up before Rio • financial mechanism for 5 conventions: CBD • UNFCCC • UNCCD • Stockholm (POPs) • Minamata
- **Convention secretariats:** Bonn — UNFCCC, UNCCD, CMS • Geneva — CITES, IPCC, Basel-Rotterdam-Stockholm • Gland — IUCN, WWF, Ramsar • Montreal — CBD
- **India-centred platforms:** ISA (2015) — Gurugram • CDRI (2019) — New Delhi • International Big Cat Alliance (2023) — India • GCF — Songdo, South Korea

Points to Note

1. TRAFFIC is **not** a UNEP bureau; it is a joint non-governmental programme of WWF and IUCN
2. Red Data Book: pink pages = critically endangered • green pages = formerly endangered, now secure species
3. IPCC does no original research — it assesses published research
4. ISA = International Solar Alliance (Gurugram) ≠ International Seabed Authority (Kingston, Jamaica)
5. CITES signed in Washington, secretariat in Geneva

One-line Facts

- **Montreux Record:** register of Ramsar sites where ecological character has changed due to human interference • only 2 Indian sites: **Keoladeo** (Rajasthan, 1990) • **Loktak** (Manipur, 1993) • Chilika removed in 2002
- **Rio+5 (1997):** New York
- **Eco-Mark:** label for eco-friendly consumer products • BIS, since 1991 • logo **earthen pot** • covers soaps, paper, paints, food items; not eggs or pesticides
- **Agmark:** purity mark for agricultural produce • Agricultural Produce (Grading and Marking) Act **1937**
- **Sikkim:** world's first 100% organic state (2016) • first 100% organic Union Territory **Lakshadweep**

Points to Note

1. Montreux Record (Ramsar) ≠ Montreal Protocol (ozone)

10 Major Environmental Laws & Regulatory Bodies of India

PYQ 21×

Major Environmental Acts: Chronology & Provisions 21×

- **Constitutional basis:** 42nd Amendment 1976 • Article **48A** (Directive Principle) • **51A(g)** (Fundamental Duty) • Forests & wildlife moved from State List to Concurrent List
- **Wildlife (Protection) Act 1972:** declaration of National Parks & Wildlife Sanctuaries • ban on hunting scheduled species • 2002 amendment — Conservation & Community Reserves
- **Wildlife (Protection) Amendment Act 2022:** schedules **6 → 4** • Schedule I highest protection • II lesser protection • III protected plants • IV CITES-listed species • **Vermin schedule removed** • control of invasive alien species
- **Water (Prevention & Control of Pollution) Act 1974: Article 252** (on resolutions of State legislatures) • under it, **Central Pollution Control Board (CPCB)** & State Boards set up in Sept 1974
- **Forest (Conservation) Act 1980:** non-forest use or de-reservation of forest land only with **prior approval of the Centre**
- **Forest Amendment Act 2023:** renamed **Van (Sanrakshan Evam Samvardhan) Adhiniyam, 1980** • in force 1 Dec 2023 • applies only to notified forests or land recorded as forest in government records on or after 25 Oct 1980 • strategic linear projects within 100 km of border/LoC/LAC exempted • zoos, safaris, eco-tourism = forestry activities
- **Air (Prevention & Control of Pollution) Act 1981: Article 253** (to implement Stockholm 1972 decisions) • air quality standards • **1987 amendment** added 'noise' as an air pollutant
- **Environment (Protection) Act 1986:** after the **Bhopal Gas Tragedy** (2–3 Dec 1984, methyl isocyanate) • **'Umbrella Legislation'** • basis for **EIA** (1994, new 2006) • CRZ (1991 → 2011 → 2019) • Eco-Sensitive Zones (ESZ) • Noise Pollution Rules 2000 • Wetlands Rules 2017
- **Biological Diversity Act 2002:** fulfils objectives of **CBD (1992)** • **three-tier structure** — National Biodiversity Authority (**Chennai, 2003**) • State Biodiversity Boards • local Biodiversity Management Committees (People's Biodiversity Register)
- **Biological Diversity (Amendment) Act 2023:** decriminalisation (fines of ₹1 lakh–₹50 lakh instead of jail) • exemption for AYUSH practitioners & codified traditional knowledge • promotes cultivation of medicinal plants

- **Forest Rights Act 2006:** Scheduled Tribes & Other Traditional Forest Dwellers • individual & community forest rights • occupation before **13 Dec 2005** • OTFDs need an extra **3 generations (75 years)** residence • **Gram Sabha** is the initiating unit • nodal — Ministry of Tribal Affairs
- **National Green Tribunal (NGT) Act 2010:** set up **18 Oct 2010** • inspired by **Article 21** (right to a healthy environment) • not bound by Code of Civil Procedure 1908 — principles of natural justice • disposal within **6 months**

Chronology of Environmental Laws

1972	**Wildlife (Protection) Act**
1974	Water (Prevention) Act → CPCB
1977	**Water Cess Act**
1980	Forest (Conservation) Act
1981	**Air (Prevention) Act**
1986	Environment (Protection) Act — umbrella
1991	**Public Liability Insurance Act**
2002	Biological Diversity Act → NBA (Chennai)
2006	**Forest Rights Act**
2010	National Green Tribunal Act
2016	**Compensatory Afforestation Fund Act**
2022	WPA amendment: schedules 6 → 4
2023	**Van (Sanrakshan Evam Samvardhan) Adhiniyam; BD amendment**

Wildlife (Protection) Act: Schedules after 2022

Schedule I Highest protection · strictest penalty tiger, elephant, rhino, snow leopard...	**Schedule II** Species with lesser protection lower penalty than Sch. I
Schedule III Protected plant species (old VI)	**Schedule IV** Specimens listed in CITES Appendices (new schedule)

1972: 6 schedules → 2022 amendment: 4 · vermin schedule removed

Points to Note

1. Chronology: Wildlife 1972 comes first • Forest 1980 before, Air 1981 after
2. Now only 4 schedules (2022) • 'Noise' not in the original 1981 Act, added by 1987 amendment
3. Water Act under Article 252 • Air & Environment Acts under Article 253
4. Forest Rights Act: ministry is Tribal Affairs • 75-year condition does not apply to Scheduled Tribes
5. Biological Diversity Act 2002 = for CBD 1992 — Nagoya Protocol (2010) came later

Major Regulatory & Statutory Bodies

- **Central Pollution Control Board (CPCB):** statutory • under **Water Act 1974** • Air Act functions also from 1981 • HQ Delhi
- **National Green Tribunal (NGT):** principal bench **New Delhi** • zonal benches — Bhopal, Pune, Kolkata, Chennai • **third country** with a specialised environmental tribunal (after Australia, New Zealand) • appeal to Supreme Court within 90 days
- **NGT jurisdiction (7 laws):** Water 1974 • Water Cess 1977 • Forest 1980 • Air 1981 • Environment 1986 • Public Liability Insurance 1991 • Biological Diversity 2002
- **National Board for Wildlife (NBWL):** under Wildlife Act (2002 amendment) • chaired by **Prime Minister** • predecessor Indian Board for Wildlife (1952)
- **National Tiger Conservation Authority (NTCA):** by 2006 amendment of Wildlife Act • chaired by **Union Environment Minister**
- **National Ganga Council (2016):** under **Environment Act 1986** • chaired by **Prime Minister** • predecessor National Ganga River Basin Authority (2009)
- **Commission for Air Quality Management (CAQM):** statutory commission for NCR & adjoining areas (2021)

Points to Note

1. Outside NGT: Wildlife Act 1972 • Forest Rights Act 2006 • Indian Forest Act 1927
2. National Ganga Council under Environment Act 1986 — not under Water Act 1974
3. NBWL & Ganga Council chaired by PM • NTCA chaired by Environment Minister

One-line Facts

- **Animal Welfare Board of India:** set up 1962 under **Prevention of Cruelty to Animals Act 1960** • efforts of **Rukmini Devi Arundale** • HQ **Ballabhgarh (Haryana)**, earlier Chennai
- **Indian Forest Act 1927:** 3 categories of forests — Reserved, Protected, Village forests
- **Wildlife Crime Control Bureau (WCCB):** under Wildlife Act (2006 amendment), 2007 • HQ New Delhi

11 Environmental Institutes, Movements, Personalities & Important Days

PYQ 15×

Major Environmental & Research Institutes 11×

- **Forest Survey of India (FSI): Dehradun** • 1981 • biennial **'India State of Forest Report'** since 1987
- **Forest Research Institute (FRI):** Dehradun • 1906
- **Wildlife Institute of India (WII):** Dehradun • 1982
- **Indian Council of Forestry Research & Education (ICFRE):** Dehradun • 1986
- **Indian Institute of Forest Management (IIFM): Bhopal** • 1982
- **National Environmental Engineering Research Institute (NEERI): Nagpur** • 1958 • under CSIR • 5 zonal labs — Chennai, Delhi, Hyderabad, Kolkata, Mumbai 7×
- **Botanical Survey of India (BSI): Kolkata** • 1890
- **Zoological Survey of India (ZSI):** Kolkata • 1916
- **Central Arid Zone Research Institute (CAZRI) & Arid Forest Research Institute (AFRI): Jodhpur** • CAZRI 1959 • AFRI 1985
- **National Research Centre for Agroforestry & Indian Grassland and Fodder Research Institute: Jhansi** • agroforestry 1988 (now CAFRI) • grassland 1962 5×
- **Salim Ali Centre for Ornithology & Natural History (SACON): Coimbatore (Anaikatti)** • 1990
- **National Biodiversity Authority (NBA): Chennai** • 2003
- **Central Avian Research Institute (CARI): Izatnagar (Bareilly)** • 1979
- **Lucknow:** National Botanical Research Institute (NBRI) 1953 • Central Drug Research Institute (CDRI) 1951 • Indian Institute of Toxicology Research (IITR)
- **G.B. Pant National Institute of Himalayan Environment: Kosi-Katarmal, Almora** • 1988

- **Bombay Natural History Society (BNHS):** **Mumbai** • 1883 • linked with Salim Ali

Points to Note

1. FSI (1981) & FRI (1906) — both Dehradun, but different institutes
2. IIFM is in Bhopal — not Dehradun • NEERI is in Nagpur — not Delhi
3. Jhansi = grassland-fodder & agroforestry • Jodhpur = arid zone • Izatnagar = avian (CARI)
4. SACON Coimbatore, BNHS Mumbai — both linked with Salim Ali

Major Environmental Movements 8×

- **Khejarli / Bishnoi Movement (1730):** Khejarli, **Jodhpur** (Rajasthan) • **Amrita Devi Bishnoi** with 3 daughters — **363 people** martyred saving Khejri trees • against soldiers of Jodhpur's Maharaja Abhay Singh • Bishnoi sect founder **Guru Jambheshwar** (29 rules)
- **Chipko Movement (1973):** **Chamoli** in then Uttar Pradesh (now Uttarakhand) • began at **Mandal village** (Gopeshwar), April 1973 • at **Reni village** (March 1974) women led by **Gaura Devi**
- **Chipko personalities:** pioneer **Chandi Prasad Bhatt** (Dasholi Gram Swarajya Sangh • Magsaysay 1982) • **Sunderlal Bahuguna** (slogan — 'Ecology is permanent economy')
- **Silent Valley Movement (1973–1980):** **Kerala**, Palakkad • against hydroelectric project on **Kunthipuzha river**, to save evergreen forests • Kerala Sastra Sahitya Parishad & poet Sugathakumari • National Park in 1984
- **Jungle Bachao Andolan (1982):** **Singhbhum** (then Bihar, now Jharkhand) • tribals against replacing natural **Sal** forests with **Teak**
- **Appiko Movement (1983):** **Karnataka**, Uttara Kannada (Sirsi) • **Panduranga Hegde** • Kannada 'Appiko' = to hug • Chipko of the South 4×
- **Narmada Bachao Andolan (1985):** against height of **Sardar Sarovar Dam** & displacement • **Medha Patkar** • Baba Amte • Arundhati Roy
- **Navdanya Movement (1987):** **Vandana Shiva** • biodiversity, organic farming & native seed conservation

India's Major Environmental Movements



- **Maiti Movement (1994-95):** **Kalyan Singh Rawat**, Chamoli (Uttarakhand) • bride & groom plant a sapling at the bride's maternal home at marriage 2×
- **Anti-Tehri Dam Movement:** against Tehri Dam on Bhagirathi • long fasts of **Sunderlal Bahuguna** (1990s)

Points to Note

1. Chipko 1973 was in then Uttar Pradesh — Uttarakhand formed in 2000
2. Chipko pioneer Chandi Prasad Bhatt • propagator Sunderlal Bahuguna • heroine of Reni Gaura Devi
3. Appiko = Karnataka • Silent Valley = Kerala — do not swap
4. Jungle Bachao opposed planting Teak — to save Sal
5. Khejarli martyr Amrita Devi • sect founder Guru Jambheshwar

Major Environmental Personalities

- **Waterman of India:** **Rajendra Singh** • Tarun Bharat Sangh (Alwar) — johad building • Magsaysay 2001 • Stockholm Water Prize 2015
- **Birdman of India:** **Dr. Salim Ali** • autobiography 'The Fall of a Sparrow'
- **Tiger Man of India:** **Kailash Sankhala** • first Director of Project Tiger
- **Forest Man of India:** **Jadav Payeng** • single-handedly planted ~1,360-acre 'Molai forest' on **Majuli island**, Brahmaputra, Assam
- **Ice Man of India:** **Chewang Norphel** • **artificial glaciers** in Ladakh
- **Dolphin Man of India:** **Ravindra Kumar Sinha** • Gangetic dolphin conservation

Points to Note

1. Artificial glacier = Chewang Norphel • 'Ice Stupa' = Sonam Wangchuk — both Ladakh
2. Tiger Man Kailash Sankhala • Birdman Salim Ali • Waterman Rajendra Singh

Important Environment Days

- **2 February:** World Wetlands Day (Ramsar 1971) • 2026 theme — 'Wetlands and traditional knowledge: Celebrating cultural heritage' • 2025 — 'Protecting Wetlands for Our Common Future'
- **3 March:** World Wildlife Day (CITES signed 1973)
- **20 March:** World Sparrow Day
- **21 March:** International Day of Forests
- **22 March:** World Water Day
- **22 April:** Earth Day • since 1970
- **22 May:** International Day for Biological Diversity
- **31 May:** World No Tobacco Day
- **5 June:** **World Environment Day** • marks Stockholm Conference (1972) • first theme 'Only One Earth' (1974)
- **8 June:** World Oceans Day
- **17 June:** World Day to Combat Desertification & Drought
- **1–7 July:** Van Mahotsav • since 1950, K.M. Munshi
- **29 July:** International Tiger Day • from 2010 St. Petersburg Tiger Summit
- **10 August:** World Lion Day • World Biofuel Day
- **12 August:** World Elephant Day
- **16 September:** International Day for Preservation of the Ozone Layer (Montreal Protocol 1987)
- **2–8 October:** Wildlife Week (India)

- **5 October:** National Dolphin Day (Gangetic dolphin National Aquatic Animal, 2009)
- **19 November:** World Toilet Day
- **26 November:** National Milk Day
- **2 December:** **National Pollution Control Day** (marks Bhopal Gas Tragedy)
- **5 December:** World Soil Day

Points to Note

1. 22 March Water • 22 April Earth • 22 May Biodiversity — 21 March Forests, 20 March Sparrow
2. 5 June Environment (Stockholm 1972) ≠ 16 September Ozone (Montreal 1987)
3. 29 July Tiger • 10 August Lion • 12 August Elephant • 5 October Dolphin

One-line Facts

- **'Silent Spring' (1962):** **Rachel Carson** (USA) • exposed harm of pesticides like DDT • foundation of modern environmental movement 3×
- **Western Ghats committees:** **Gadgil (WGEEP, 2011)** — entire Western Ghats as Ecologically Sensitive Area (ESA) • **Kasturirangan (2013)** — about 37% as ESA 4×
- **Environment awards:** **Tyler Prize** — 'Nobel Prize for the Environment' • UNEP's 'Global 500' & 'Champions of the Earth' 5×
- **Ecomark:** launched 1991 by the Environment Ministry • standards & certification by **Bureau of Indian Standards (BIS)** • eco-friendly consumer products • logo **'earthen pot'** • Ecomark Rules 2024 6×
- **Headquarters:** **Greenpeace** (1971, Canada) — Amsterdam • **WWF** (1961) — Gland (Switzerland) • **UNEP** (5 June 1972) — Nairobi (Kenya) 7×

Points to Note

1. Tyler = 'Nobel for the Environment' • Goldman = 'Green Nobel'
2. WWF & IUCN both in Gland • Greenpeace founded in Canada, HQ Amsterdam

Part 5: Uttar Pradesh Special

12 Uttar Pradesh: Forests, National Park, Tiger Reserves & Wildlife Sanctuaries

PYQ 20×

Forest & Tree Cover, Forestry Facts & State Symbols

- **Geographical area:** 2,40,928 sq km • about 7.33% of India • 4th largest state by area
- **Forest cover (ISFR 2023):** about 15,000 sq km • about **6.2%** of geographical area
- **Tree cover:** 8,950.92 sq km (about 3.7%) • **3rd** in India by tree cover area (Maharashtra › Rajasthan › Uttar Pradesh)
- **Forest + tree cover:** about 24,000 sq km • about **10%** (national 25.17%; target 33%)
- **Increase (2021–23):** forest + tree cover +559 sq km • 2nd in India, jointly with Odisha (1st Chhattisgarh 684)
- **Districts with maximum forest cover area:** **Sonbhadra** › Lakhimpur Kheri › Mirzapur › Pilibhit › Chitrakoot 5×
- **Districts with maximum forest cover %:** **Sonbhadra** (about 35%) › Chandauli › Chitrakoot › Pilibhit
- **District with minimum forest cover:** **Bhadohi (Sant Ravidas Nagar)** • lowest by area (about 5 sq km), then Mau, Sambhal • by % also about 0.5% (nearly equal to Sambhal)
- **Tropical moist deciduous forests:** Terai-Bhabar region • Sal, Shisham, Semal, Teak, Khair
- **Tropical dry deciduous forests:** central & eastern plains, Vindhya-Bundelkhand plateau • Tendu, Palash, Mahua
- **Tropical thorn forests:** dry south-west & Bundelkhand • Babool, Khair, Kareel
- **Montane/coniferous forests:** none in UP after the formation of Uttarakhand (9 November 2000)
- **State forest policy:** first 1952 • second 1998 • current State Forest Policy 2017
- **Operation Green:** to increase plantation & green cover • launched **1 July 2001**
- **Plantation record:** 11 July 2016 • about 5 crore saplings in 24 hours • Guinness World Record
- **State symbols:** animal **Barasingha** (swamp deer, Rucervus duvaucelii) • bird **Sarus crane** (Kraunch) • tree Ashoka (Sita Ashoka) • flower Palash (Tesu, 'Flame of the Forest') • aquatic animal Gangetic dolphin (17 October 2023) 4×

State Symbols of Uttar Pradesh

State animal	State bird	Aquatic animal	State tree	State flower
Barasingha (swamp deer)	Sarus crane tallest flier	Ganga dolphin (17 Oct 2023)	Ashoka	Palash "forest flame"

Forest cover ≈ 6.2% (ISFR 2023) · tree cover 8,950.92 sq km (3rd in India)

Points to Note

1. State animal is Barasingha (swamp deer) — the tiger is the national animal, not the state animal
2. Sarus = world's tallest flying bird; 'Kraunch' is its name — the peacock is the national bird
3. Sonbhadra is 1st by both area and %; Lakhimpur Kheri is 2nd only by area
4. Forest cover (about 6.2%) and forest + tree cover (about 10%) are different figures — do not mix them up in options
5. Standard answer for minimum forest cover is Bhadohi (Sant Ravidas Nagar); Sambhal is nearly equal by %

National Park, Tiger Reserves & Elephant Reserves

16×

- **Only national park:** **Dudhwa** (Lakhimpur Kheri) • India–Nepal border • about 490 sq km
- **Dudhwa timeline:** 1958 wildlife sanctuary (for Barasingha) → **1 February 1977** national park (efforts of Billy Arjan Singh) → 1987 Kishanpur added to form Dudhwa Tiger Reserve (Project Tiger) → Katarniaghat added later (2000)
- **Key fauna of Dudhwa:** tiger • Barasingha • one-horned rhinoceros • elephant • Bengal florican
- **Rhino rehabilitation project:** 1984 • rhinos brought from Assam & Nepal • second rhino rehabilitation area also in Dudhwa
- **Rivers of Dudhwa:** Suheli • Mohana (on the Nepal border)
- **UP's 4 tiger reserves (order):** Dudhwa 1987 → Amangarh 2012 → Pilibhit 2014 → Ranipur 2022 4×

- **Dudhwa Tiger Reserve (1987):** Lakhimpur Kheri + Bahraich • state's first & largest • Dudhwa National Park + Kishanpur (1987) + Katarniaghat sanctuary (added 2000)
- **Amangarh Tiger Reserve (2012):** **Bijnor** • buffer area of Jim Corbett Tiger Reserve (Uttarakhand), declared a tiger reserve in UP's part
- **Pilibhit Tiger Reserve (2014):** Pilibhit + Shahjahanpur • tigers doubled in 2014–18 → first international **TX2 Award (2020)** • source of the Gomti (Fulhar Jheel, Madhotanda) nearby
- **Ranipur Tiger Reserve (2022):** **Chitrakoot** • UP's 4th, India's **53rd** • first in Bundelkhand • corridor with Panna Tiger Reserve (Madhya Pradesh)
- **Tiger count (2022 estimation):** UP 205 (2018: 173)
- **Uttar Pradesh (Shivalik) Elephant Reserve:** Saharanpur–Bijnor • 2009 • state's first
- **Terai Elephant Reserve (TER):** Dudhwa–Pilibhit landscape • approved 2022 • India's **33rd**, UP's second • about 3,049 sq km

Points to Note

1. UP has only 1 national park (Dudhwa) but 4 tiger reserves — Pilibhit, Amangarh, Ranipur are not national parks
2. TX2 Award went to Pilibhit Tiger Reserve — not Dudhwa
3. Amangarh (Bijnor) is Corbett's buffer; Corbett itself is in Uttarakhand
4. Ranipur: wildlife sanctuary 1977, tiger reserve 2022 — India's 53rd, but the current total of tiger reserves is 58
5. Katarniaghat is in Bahraich, Kishanpur in Lakhimpur Kheri — both are part of Dudhwa Tiger Reserve

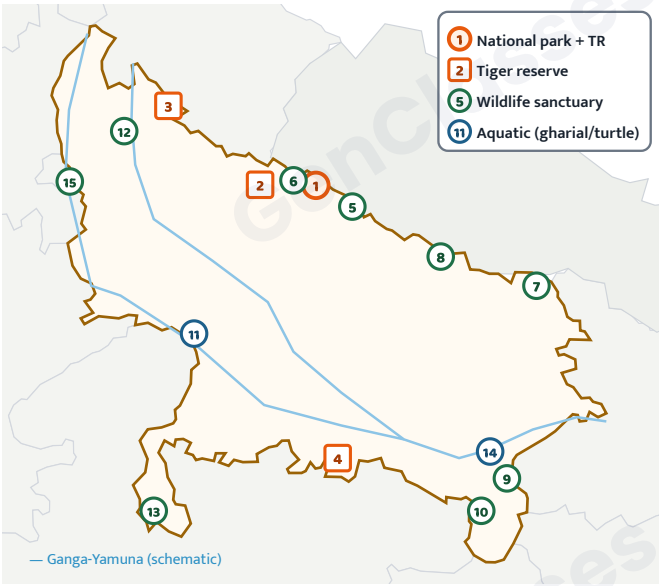
Wildlife Sanctuaries

- **Oldest (first):** **Chandraprabha** (Chandauli) • 1957 • Asiatic lions released in 1958 (experiment failed) • Rajdari-Devdari waterfalls 4×
- **Largest:** **Hastinapur** • 1986 • 2,073 sq km • Meerut, Muzaffarnagar, Hapur, Bijnor, Amroha • Haiderpur wetland lies within it
- **Smallest:** **Mahavir Swami** (Lalitpur) • 1977 • 5.41 sq km
- **Kishanpur:** Lakhimpur Kheri • 1972 • part of Dudhwa Tiger Reserve
- **Katarniaghat:** Bahraich • 1975 • Girwa (Geruwa) river — gharial, Gangetic dolphin • part of Dudhwa Tiger Reserve
- **Ranipur:** Chitrakoot (earlier Banda) • 1977 • now part of Ranipur Tiger Reserve
- **National Chambal:** Agra, Etawah • 1979 • joint project of UP–MP–Rajasthan • gharial, dolphin
- **Kaimoor:** Mirzapur, Sonbhadra • 1982 • blackbuck, chinkara
- **Sohagibarwa:** Maharajganj • 1987 • Nepal border
- **Sohelwa:** Balrampur, Shravasti (old Gonda region) • 1988 • Nepal border
- **Turtle (Kashi) Wildlife Sanctuary:** Varanasi • 1989 • now shifted to the Ganga stretch between Prayagraj–Mirzapur–Bhadohi
- **Pilibhit:** Pilibhit • 2014 • now part of Pilibhit Tiger Reserve

Points to Note

1. Sohagibarwa (Maharajganj) ≠ Sohelwa (Balrampur–Shravasti) — similar names
2. Largest sanctuary is Hastinapur, smallest is Mahavir Swami (Lalitpur) — Lalitpur ≠ Lakhimpur
3. First wildlife sanctuary Chandraprabha (1957); first bird sanctuary Nawabganj (Unnao, 1984)
4. National Chambal Sanctuary spans three states; UP part is Agra–Etawah

Uttar Pradesh: National Park, Tiger Reserves & Key Sanctuaries



1 Dudhwa (Lakhimpur Kheri) · 2 Pilibhit · 3 Amangarh (Bijnor) · 4 Ranipur (Chitrakoot) · 5 Katarniaghat (Bahraich) · 6 Kishanpur (Lakhimpur Kheri) · 7 Sohagibarwa (Maharajganj) · 8 Suhelwa (Balrampur-Shravasti) · 9 Chandraprabha (Chandauli) · 10 Kaimoor (Mirzapur-Sonbhadra) · 11 National Chambal (Agra-Etawah) · 12 Hastinapur (Meerut) · 13 Mahavir Swami (Lalitpur) · 14 Kashi Turtle (Varanasi) · 15 Okhla (Gautam Buddh Nagar)

One-line Facts

- **UP's 3 zoological parks:** Nawab Wajid Ali Shah (Lucknow, 1921) • Kanpur (Allen Forest, 1974) • Shaheed Ashfaq Ulla Khan (Gorakhpur, 2021)
- **India's first Conservation & Community Reserve:** Conservation Reserve — Tiruppadaimarathur (Tirunelveli, Tamil Nadu) • Community Reserve — Keshopur-Miani (Gurdaspur, Punjab)

13 Uttar Pradesh: Bird Sanctuaries, Ramsar Sites & Latest Initiatives

Bird Sanctuaries 9×

- **First (oldest):** **Nawabganj (Shaheed Chandra Shekhar Azad)** • Unnao • 1984 • on Lucknow–Kanpur road • Ramsar
- **Largest:** **Lakh Bahosi** • Kannauj • about 80 sq km • two lakes, Lakh & Bahosi
- **Smallest:** **Shekha Jheel** • Aligarh • 0.40 sq km • Ramsar
- **Okhla:** Gautam Buddh Nagar (Noida) • 1990 • on the Yamuna, Delhi–UP border (Okhla barrage)
- **Sur Sarovar (Keetham Lake):** Agra • man-made reservoir beside the Yamuna • Bear Rescue Centre nearby
- **Patna:** **Etah** (Jalesar) • 1990 • 1.09 sq km 3×
- **Saman:** **Mainpuri** • 1990 • seasonal oxbow lake on the Ganga floodplain • Greylag goose, sarus
- **Sandi:** Hardoi • ancient name Dahar Jheel • near the Garra river
- **Samaspur:** Raebareli • 1987 • 6 interconnected lakes
- **Parvati Arga:** Gonda • two oxbow lakes, Parvati & Arga
- **Bakhira:** Sant Kabir Nagar (earlier Basti) • 1990 • largest natural floodplain lake of eastern UP 4×
- **Surha Tal (Loknayak Jayaprakash Narayan):** Ballia • oxbow lake in the Ganga–Ghaghara (Saryu) doab
- **Dr. Bhimrao Ambedkar:** Pratapgarh (Benti, Kunda) • 2003

- **Vijay Sagar:** Mahoba • Bundelkhand

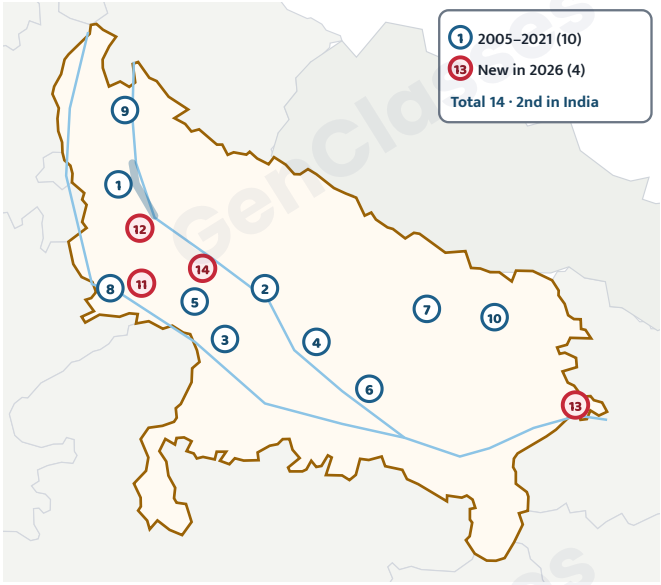
Points to Note

1. Patna Bird Sanctuary is in Etah (Jalesar) — not in Bihar
2. Bakhira's district: old answers say 'Basti', now Sant Kabir Nagar
3. Smallest is Shekha Jheel (0.40 sq km); choose Patna (1.09) only if Shekha is not among the options
4. Okhla is on the Yamuna in Gautam Buddh Nagar — not in Delhi

Ramsar Sites 11×

- **Uttar Pradesh:** 14 Ramsar sites • **2nd** in India (after Tamil Nadu 20) • India total 103 (October 2026)
- **Upper Ganga River (Brijghat–Narora stretch):** Hapur–Bulandshahr • 2005 • state's **first** & largest (26,590 ha, about 85 km) • Gangetic dolphin, gharial
- **Nawabganj Bird Sanctuary:** Unnao • 2019
- **Parvati Arga Bird Sanctuary:** Gonda • 2019
- **Saman Bird Sanctuary:** Mainpuri • 2019
- **Samaspur Bird Sanctuary:** Raebareli • 2019
- **Sandi Bird Sanctuary:** Hardoi • 2019
- **Sarsai Nawar Lake:** Etawah • 2019 • Sarus crane conservation • near Hazari Mahadev temple
- **Sur Sarovar (Keetham Lake):** Agra • 2020
- **Haiderpur Wetland:** Muzaffarnagar–Bijnor • 2021 • inside Hastinapur Sanctuary • formed by Madhya Ganga Barrage (1984)
- **Bakhira Bird Sanctuary:** Sant Kabir Nagar • 2022
- **Patna Bird Sanctuary:** Etah • January 2026 (with Chhari Dhand, Kachchh)
- **Shekha Jheel:** Aligarh • April 2026 • India's 99th • state's smallest (40.31 ha)
- **Surha Tal (Jayaprakash Narayan Bird Sanctuary):** Ballia • June 2026 (announced 5 June) • India's **100th**
- **Sauj Jheel: Mainpuri** • October 2026 • India's **103rd** (latest) • Mainpuri's second Ramsar site

Uttar Pradesh's 14 Ramsar Sites



1 Upper Ganga (Brijghat–Narora) · 2 Sandi (Hardoi) · 3 Sarsai Nawar (Etawah) · 4 Nawabganj (Unnao) · 5 Saman (Mainpuri) · 6 Samaspur (Raebareli) · 7 Parvati Arga (Gonda) · 8 Sur Sarovar (Agra) · 9 Haiderpur (Muzaffarnagar-Bijnor) · 10 Bakhira (Sant Kabir Nagar) · 11 Patna BS (Etah) · 12 Shekha Jheel (Aligarh) · 13 Surha Tal (Ballia) — 100th · 14 Sauj Jheel (Mainpuri) — 103rd

Points to Note

1. Surha Tal (Ballia, 2026) ≠ Sur Sarovar (Agra, 2020)
2. Mainpuri has two Ramsar sites — Saman (2019) & Sauj (2026)
3. State's first Ramsar site is Upper Ganga (2005) — not a bird sanctuary; it is also the largest, the smallest is Shekha Jheel
4. Sarsai Nawar is not a bird sanctuary but a lake/wetland; Haiderpur lies inside Hastinapur Sanctuary
5. The six 2019 sites were announced in January 2020 — questions may also say 2020

Latest Initiatives 7×

- **Jatayu Conservation & Breeding Centre:** Maharajganj (Farenda, Bhari-Baisi) • world's first centre for the **Asian King Vulture (Red-headed vulture)**
- **Night Safari & Biodiversity Park:** Kukrail forest area, Lucknow • India's first night safari • modelled on Singapore Night Safari
- **Dolphin conservation:** Chambal (Etawah) • Ganga (Bijnor) • first river dolphin estimation (2025): UP **2,397**, highest in India

Points to Note

1. Maharajganj = world's first King Vulture centre; Pinjore (Haryana) = India's first vulture breeding centre
2. Night Safari is at Kukrail (Lucknow), Lion Safari at Etawah — two different things
3. UP is 1st in dolphin numbers, Bihar in density; National Aquatic Animal 2009, UP State Aquatic Animal 2023

One-line Facts

- **Asiatic Lion Breeding Centre & Lion Safari:** Etawah
- **Gharial Breeding & Rehabilitation Centres:** Kukrail (Lucknow) • Katarniaghat (Bahraich)
- **Peacock Conservation Centre:** Vrindavan (Mathura)
- **Bear Rescue Centre:** Agra (Keetham/Sur Sarovar)

Join Telegram today for the next booklet
Booklets for all subjects, Extra Notes PDF and Concept Notes — daily and weekly
t.me/genclassesofficial · www.genclasses.com